UNIVERSITÀ DI PARMA

IoT Lab

→ LEC. 15
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LEC. 13 → 6LoWPAN → Dr. Alfieri
LEC. 14 → LoRaWAN → Prof. Bellini)

Routing in LLNs & RPL

Main sources

- J. P. Vasseur and A. Dunkels, `Interconnecting Smart Objects with IP,` Morgan Kaufmann, 2010. Chapter 17.

- Brandtr, Clausen, Dawson-Haggerty, Hui, Pister, Thubert, Winter, *RPL (ripple): Routing Protocol for Low Power and Lossy Networks – Walkthrough*, IETF `draft-dt-roll-rpl-01.txt`, IETF 75 – Roll WG – July 2009

From WSNs to RPL (1/2)

- Distance Vector routing (e.g., AODV) is widely used in WSNs
  - trade off between capabilities and resources
- Reduced states in nodes (vs. Link State)
- Simplicity => more control cost over time
- Improvements proposed
  - Loop detection
    - Packets received from a node that is on the way to the destination cause routing control
  - Trickle (adaptive adv period + suppression)
    - Limits the degradations due to node spatial density
  - Explicit link evaluation
    - Prior to selecting a parent as feasible

From WSNs to RPL (2/2)

- Expected Transmission Count (ETX) – MIT
  - Measure of the quality of a path between two nodes in a wireless packet data network
  - A lot more relevant than hop count in wireless
- Wikipedia:

"ETX is the number of expected transmissions of a packet necessary for it to be received without error at its destination. This number varies from 1 to infinity. An ETX of 1 indicates a perfect transmission medium, where an ETX of infinity represents a completely non-functional link. Because ETX is an *expected* transmission count for a future event, as opposed to an actual count of a past event, it represents a probability, and is therefore a real number, and not an integer. For example, if it took 1898 transmissions to transfer 1024 packets without error, the ETX on the link is 1898/1024, or approximately 1.85. Due to varying characteristics of the transmission medium, the number may vary widely."

$$ETX = \frac{\sum_{i=1}^{N} T_i}{N} \xrightarrow{N \to \infty} E[T_i] = E[T]$$

Distance Vector (DV) protocol in IP Networks



- RIP (Routing Information Protocol): #hops as metric
  - Unfit for large, unstable & complex topologies
  - Sync effect, transient loops, count to infinity
  - Partial fixes by split horizon (poison reverse)
- More recent protocols
  - Trade RIP's loops for temporary black holes
  - Path hold-down then route poisoning
  - Quarantine path for which hop counts goes up

The link model

- LLN connectivity might not match the traditional wired link model
  - Links are not a given.
  - Statistical connectivity might cause transient unreachability from/to a rather good neighbor
  - and might enable to receive a packet from far far away from an improbable neighbor
  - Not a RPL-specific issue but a LLN-issue that the IETF ROLL WG has considered



Low-power and Lossy Networks (LLNs)

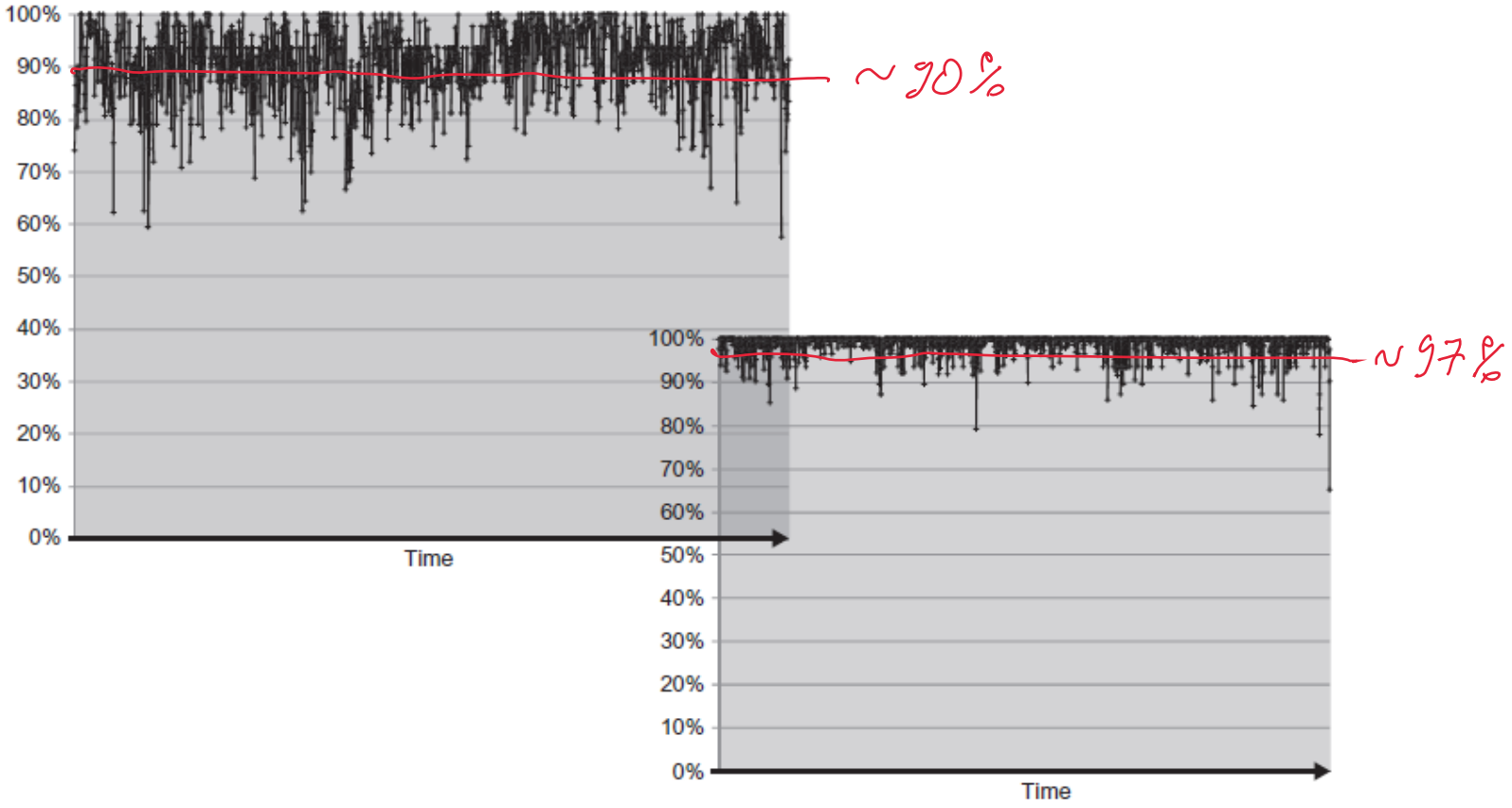


FIGURE 17.1

Packet Delivery Ratio for two IEEE 802.15.4 links.

Characterstics and Applications

- Common Characteristics
  - Low power and Lossy Networks (LLNs) consisting largely of constrained nodes.
  - Lossy and unstable links, typically supporting low data rates, relatively low packet delivery rates.
  - Traffic patterns are not simply point-to-point, but in many cases point-to-multipoint (P2MP) or multipoint-to-point (MP2P)
  - Potentially comprising up to thousands of nodes.
- Four main application scenarios
  - Urban LLNs (including smart grids) [RFC5548]
  - Industrial LLNs [RFC5673]
  - Home Automation LLNs [RFC5826]
  - Building Automation LLNs [RFC5867]

Requirements (1/2)

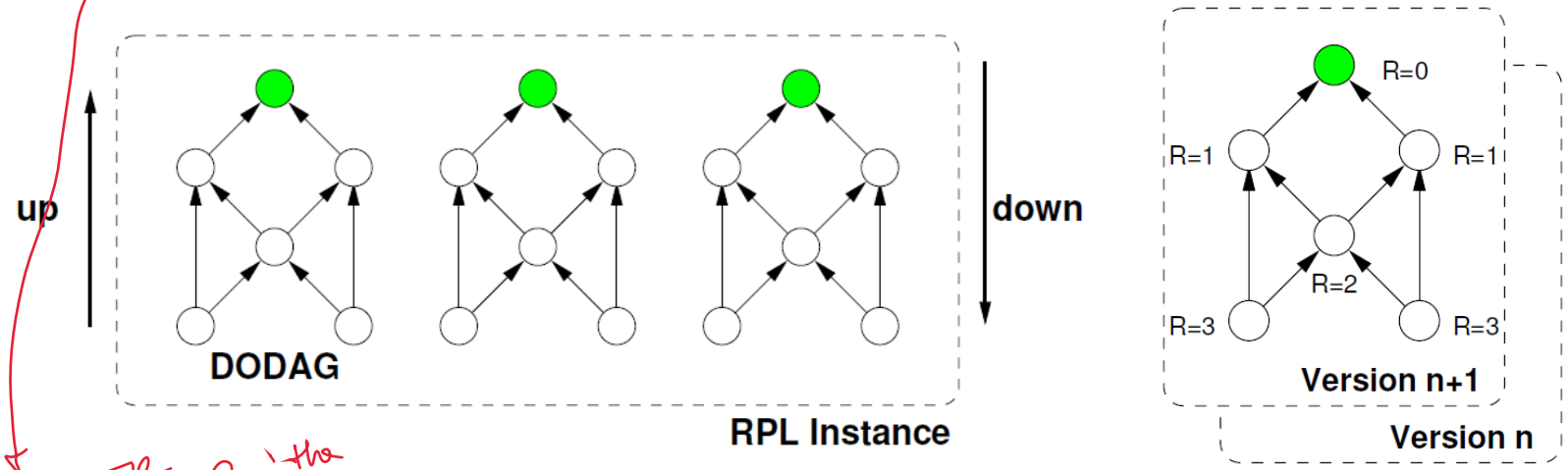
- Unicast/anycast/multicast
- Adaptive routing: new paths are dynamically and automatically recomputed as conditions change in the network (e.g., link/node failure, mobility, etc.)
- Constraint-based routing (energy, CPU, and memory as well as link attributes, such as link latency)
- Traffic characteristics (MP2P-data collection, P2MP-sw updates, P2P, computation of parallel paths)
- Scalability: from 250 to 10000 nodes
- Configuration and management: minimum or zero-configuration; isolation of a misbehaving node to limit/eliminate its impact on other nodes

Requirements (2/2)

- Node attribute: with sleeping nodes in the network (a frequent situation with battery-operated nodes), the routing protocol must discover the capability of a node to act as a proxy
- Performance: ‘‘tricky’’ requirement (e.g., a routing protocol must provide mobility with a convergence time below 0.5 s and it must converge within 0.5 s if no nodes have moved and within 2 s if the destination has moved)
- Security:
  - wide range (low to high);
  - the routing protocol must gracefully handle routing temporal security updates (e.g., dynamic keys) to sleeping devices on their ‘ *awake* ’ cycle to assure that sleeping devices can readily and efficiently access the network

RPL Approach (1/2)

- Design a **modular** routing protocol where the core component of the application would be specified by the RPL specification with optional features activated only where and when needed.
- For example, RPL specifies how to build a destination oriented directed acyclic graph (DODAG), but the characteristics of the DODAG are specified by an objective function.



RPL Approach (2/2)

- For the time being, think of a DODAG as a logical routing topology over a physical network that is built by the routing protocol to meet specific criteria.
- A node may join multiple DODAGs (if the application requires different objectives that must be realized through the use of multiple DODAGs) and mark the traffic according to the DODAG characteristics in support of Quality of Service (QoS) awareness and constrained-based routing.
- DODAG Properties
  - Many-to-one communication (MP2P): upwards
  - One-to-many communication (P2MP): downwards
  - Point-to-point communication (P2P): upwards-downwards

Routing Metrics in IP Nets

- Most of the IP routing protocols used in today's networks such as OSPF use static link metrics.
  - The network administrator is responsible for configuring the link metrics, which may refl ect the link bandwidth, delay, or combine several metrics.
  - Some Service Providers are combining up to three metrics (e.g., delays, bandwidth, cost) in the link metric.
  - Then the routing protocol computes the shortest path taking into account these static link metrics.
- Dynamic link metrics.
  - In ARPANET-2: studies to dynamically compute the link metric based on the averaged queue length to reflect the level of congestion.
  - These strategies were abandoned due to the difficulty in designing stable systems.
  - One of the main challenges with dynamic metrics is to carefully control the rate at which new metrics are advertised. Frequent link metric refreshers provide a high level of accuracy but may also lead to routing oscillation.

Routing in LLNs



- *Metric*: a scalar used to determine the best path according to some objective function (OF).
  - For example, if the link metric is representative of the link propagation delay, the path cost represents the total propagation delay to the destination and the OF may specify ''finding the shortest path based on the propagation delay.''
  - Some metrics may not be additive; for example, the OF may be ''finding the path where the minimum link quality is maximized. ''
- *Constraint*: used to include or eliminate links or nodes that do not meet specific criteria (this is usually referred to as *constraint-based routing*).
  - For example, the OF may not select any path that traverses a node that is battery-operated or a link that does not provide link layer encryption.
- The OF may combine link/node metrics and constraints such as " find the path with the minimum delay that does not traverse any non-encrypted link. "

Objective Function (OF)

- Examples (IP nets):
  - the OF of RIP is to select the path with minimal hop count;
  - OSPF or IS-IS would compute the paths that provide the minimum cost where the path cost is simply the sum of the static link cost along the path.
  - With MPLS TE the OF may be slightly more complex: " find the shortest path according to some metric such as the OSPF/ IS-IS
  - metric or the Traffic Engineering metric that satisfies some constraint such as the available reservable bandwidth or the type of recovery protection provided by the link. "
- LLNs: strong interest in using several OFs because deployments greatly vary with different objectives and a single network may support traffic with very different requirements in terms of path quality.
  - Consider the case of a mixed network with battery- and main-powered nodes, a variety of high and low bandwidth links, and two main applications (telemetry and critical alarms).
  - Extremely useful for each node: supporting both applications to be able to use two paths.
    - One " time-sensitive " path for alarms where the objective is to have a short delay and a highly reliable path with no constraint on the type of nodes along the path to the destination
    - Another " not time-sensitive path " for the telemetry traffic where it is beneficial to not traverse any battery-operated node to preserve energy and where the objective would be to minimize hops to avoid traffic congestion in the network.
  - RPL addresses these requirements by building two DODAGs with each one having its own OF. The OF is used in conjunction with the routing metric to compute the path.

Routing Metrics (1/2)



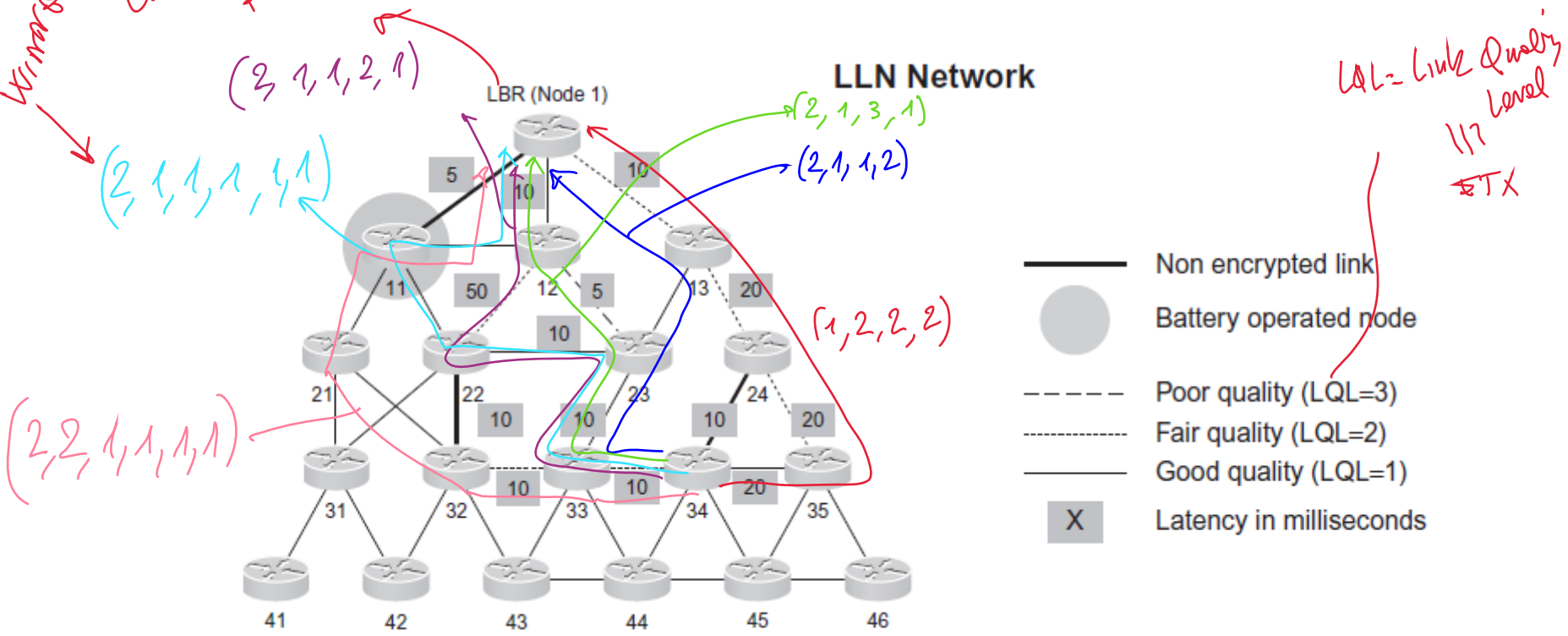
- *Aggregated* (path cost is defi ned as the sum of the cost of all links along the path) versus *Recorded* (record each individual link metric)
- *Local* (not propagated along the DODAG) versus *Global*
- The Routing Metrics/Constraints Common Header (several flags → routing object: refers to a routing metric or a constraint; is local versus global; if the global metric is aggregated versus recorded; if a constraint is optional or mandatory, and if a metric is additive or reports a maximum/minimum.)
- Routing objects
  - The *Node State and Attributes (NSA) Object*: nodes may act as traffic aggregators (use many bits to describe the state); nodes have limited available resources → 1-bit fl ag set when a node sustainably experiences some level of congestion
  - *Node Energy Object*: several levels of granularity to characterize the node energy: (1) the node power mode; (2) the estimated remaining lifetime; and (3) potentially some detailed set of power-related metrics and attributes.
  - *Hop-count Object*
  - *Throughput Object*
  - *Latency Object*

Routing Metrics (2/2)



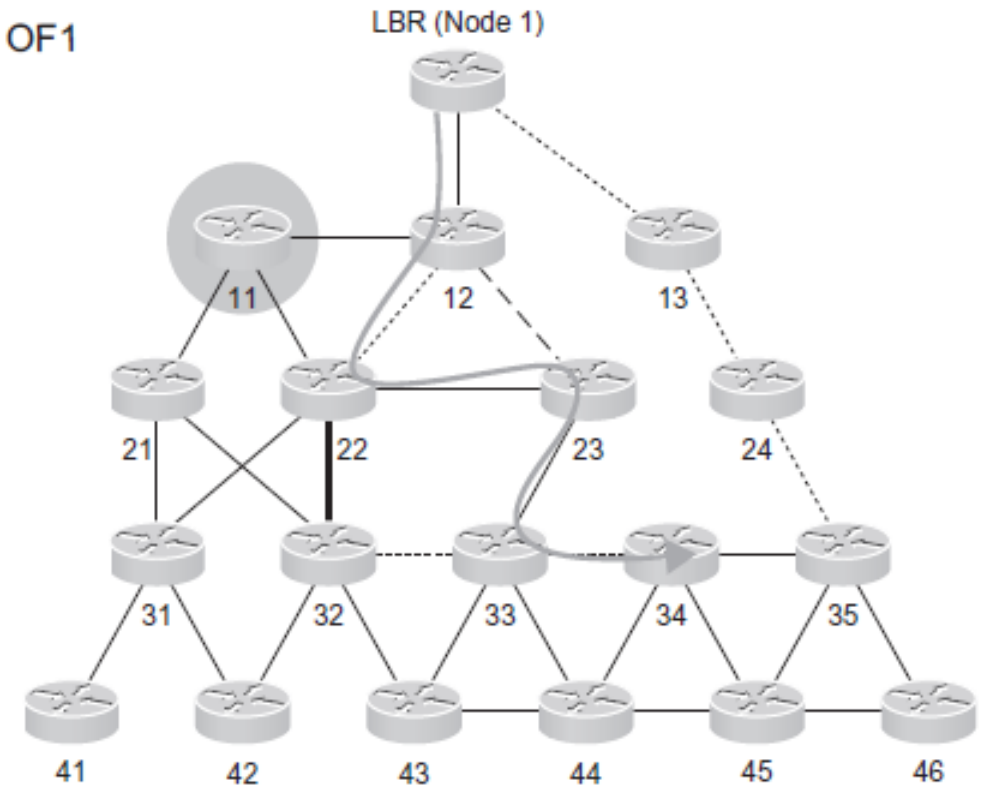
- Link Reliability Object: the most popular reliability metric thus far is the expected transmission (ETX) count metric
  - Characterizes the average number of packet transmissions required to successfully transmit a packet (see before).
  - Tightly coupled to the throughput along a path.
  - Several techniques have been proposed to compute ETX. → IETF does not specify the method for computing ETX values. The ETX is a link-specific quantity and the technique used to compute the ETX value should be independent of the link layer and not specified by the network layer that only carries it for routing protocol decisions.
- Link Colors Attribute: link color is encoded using a bit vector; the meaning of each color is left to the implementer.
  - RPL computes paths over a dynamically built DODAG. The DODAG root uses an OF required for each node along the path reporting path metrics to also report the set of colors of each link along the path.
  - For example, suppose that the color blue is used to indicate the support of the link layer encryption. Upon receiving the path metric, if link colors are recorded, a node may decide to elect as a parent the parent reporting paths with encrypted links (blue links) or with the maximum number of blue links in the absence of a path exclusively made of blue links.

An Example (1/3)



- Routing: from the low-power and lossy network border router (LBR) to node 34
- OF1: " Use the LQL as a global recorded metric and favor paths with the minimum number of low and fair quality links, use the link color as a link constraint to avoid non-encrypted links. "
- OF2: " Find the best path in terms of latency (link latency is used as a global aggregated metric), while avoiding poor quality links and battery-operated nodes. "

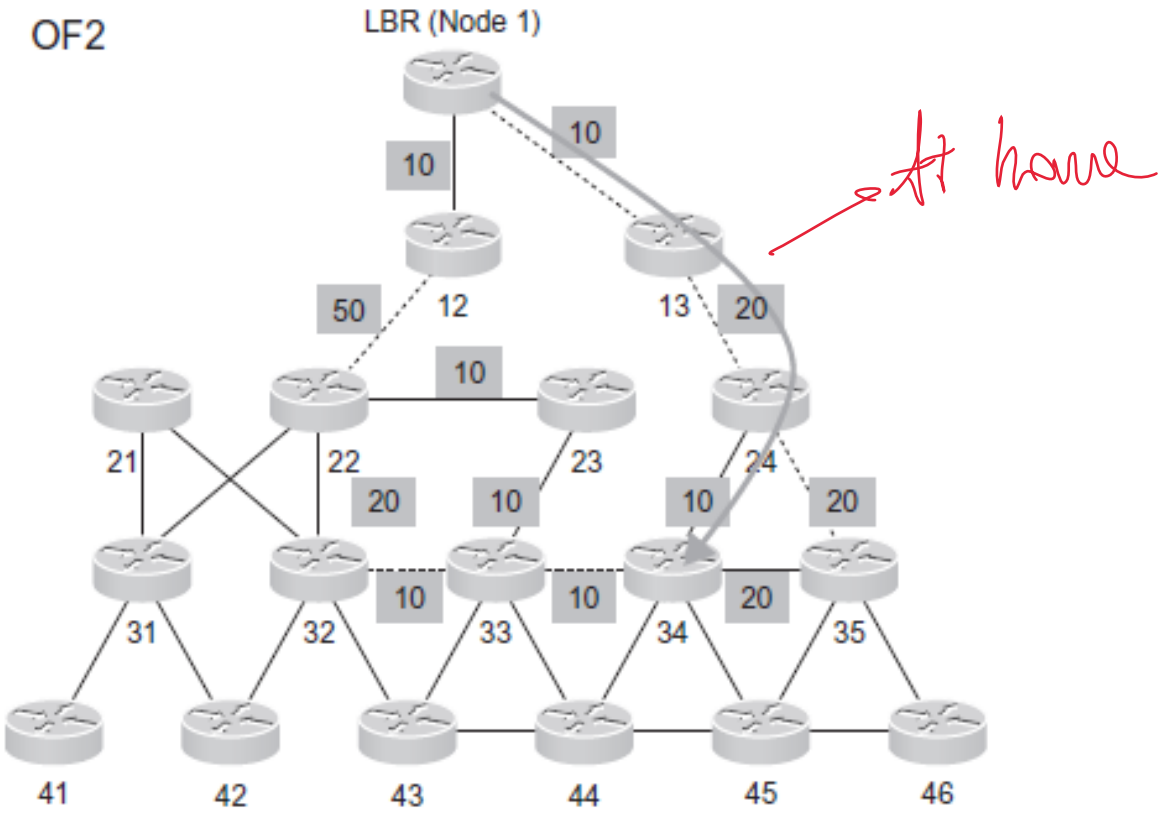
An Example (2/3)



(1,2,2,2) (2,1,1,2,1)

- OF1: " Use the LQL as a global recorded metric and favor paths with the minimum number of low and fair quality links, use the link color as a link constraint to avoid non-encrypted links. "
- Two paths are available with an equivalent aggregated LQL metric: 34-35-24-13-1 and 34-33-23-22-12-1. Since OF specifies using a recorded metric → path 34-33-**23-22**-12-1 is chosen since it only has two links of " fair " quality. → no! Check the

An Example (3/3)



- OF2: " Find the best path in terms of latency (link latency is used as a global aggregated metric), while avoiding poor quality links and battery-operated nodes. "
- Several paths have been pruned because they traverse battery-operated nodes (node 11) and traverse poor quality links (link 12-23). The best path (lowest latency) is 34-24-13-1.

RPL Basics





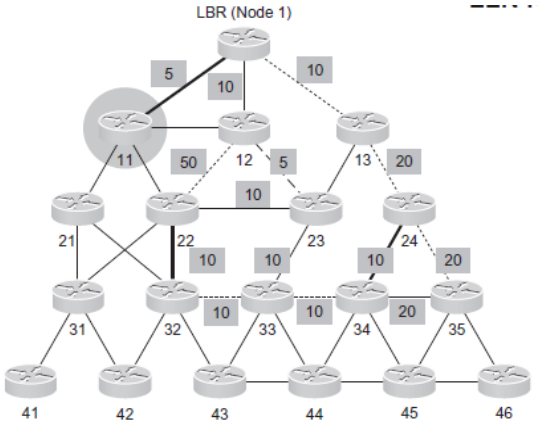
- RPL was designed to be **highly modular**: the core of the routing protocol would address the intersection of the application-specific routing requirements, and additional modules would be added as needed to address specific requirements.
- RPL was designed for LLNs where constrained devices are interconnected by (wireless and wired) lossy links → a node should carefully observe a link and start using it or determine whether to stop using it (thus triggering a global path recomputation in the network).
- Control traffic must be as tightly bounded as possible, ideally the control traffic should decrease as the routing topology stabilizes
- RPL is a distance vector protocol that builds a DODAG where paths are constructed from each node in the network to the DODAG root (typically a sink or an LBR)
- In contrast with tree topologies, DODAGs offer redundant paths, which is a MUST requirement for LLNs. Thus if the topology permits, RPL may provision more than one path between a node and the DODAG root and even other nodes in the network.

RPL Operations (1/5)

- First, one or more nodes are configured as DODAG roots by the network administrator.
- A node discovery mechanism based on newly defined ICMPv6 messages is used by RPL to build the DODAG.
- RPL defines two new ICMPv6 messages:
  - DODAG information object (DIO) messages; → UP & DOWN
  - destination advertisement object (DAO) messages. → ~~DOWN~~ UP
- DIO messages (simply referred to as DIO) are sent by nodes to advertise information about the DODAG, such as: the DODAGID; the OF; the DODAG rank (depth); the DODAGSequenceNumber; along with other DODAG parameters such as a set of path metrics and constraints discussed before.
- When a node discovers multiple DODAG neighbors (that could become parents or sibling), it makes use of various rules to decide whether (and where) to join the DODAG. This allows the construction of the DODAG as nodes join.
- Once a node has joined a DODAG, it has a route toward the DODAG root (which may be a default route) in support of the MP2P traffic from the leaves to the DODAG root (in the up direction).

RPL Operations (2/5)

- RPL uses “ up ” and “ down ” directions terminology.
  - up direction from a leaf toward the DODAG root;
  - down direction refers to the opposite direction.
- The usual terminology of parents/children is used.
- RPL also introduces the “sibling”: two nodes are siblings if they have the same rank in the DODAG (note that they may or may not have a common parent).
- The parent of a node in the DODAG is the immediate successor within the DODAG in the up direction, whereas a DODAG sibling refers to a node at the same rank.
- In the side figure, 13 is a parent of 24, 22, and 23 are siblings, and 43 and 44 are children of 33.
- A DODAG is said to be:
  - *grounded* if it is connected to what RPL calls a “ goal, ” which can be a node connected to an external (non-LLN) private IP network or the public Internet;
  - a non-grounded DODAG is called a *floating* DODAG.



- RPL uses iterations controlled by the DODAG root to maintain the DODAG; the DODAGSequenceNumber is a counter incremented by the DODAG root to specify the iteration number of the DODAG .

RPL Operations (3/5)

- DAO messages: provide routing information in the down direction (for the traffic from the route to the leaf) and for the P2P direction since the DODAG provides defaults routes to the DODAG root from each node in the network.
- DAO messages (simply referred to as DAO) are used to advertise prefi x reachability toward the leaves. DAOs carry prefix information along with a lifetime (to determine the freshness of the destination advertisement-DAs) and depth or path cost information to determine how far the destination is.
- Note that the path in the downward direction is dictated by the DODAG built by RPL in the other direction.
- In some cases, DAOs may also record the set of visited nodes → particularly useful when the intermediate nodes cannot store any routing states.
- If a parent receives destination advertisements that can be aggregated from multiple children, local policy may be used to perform prefix aggregation in an attempt to reduce routing table and the size of DAO messages.
- Note that redundant DAO messages are aggregated along the DODAG. An OF may be specifically designed to maximize prefix aggregation.

RPL Operations (4/5)

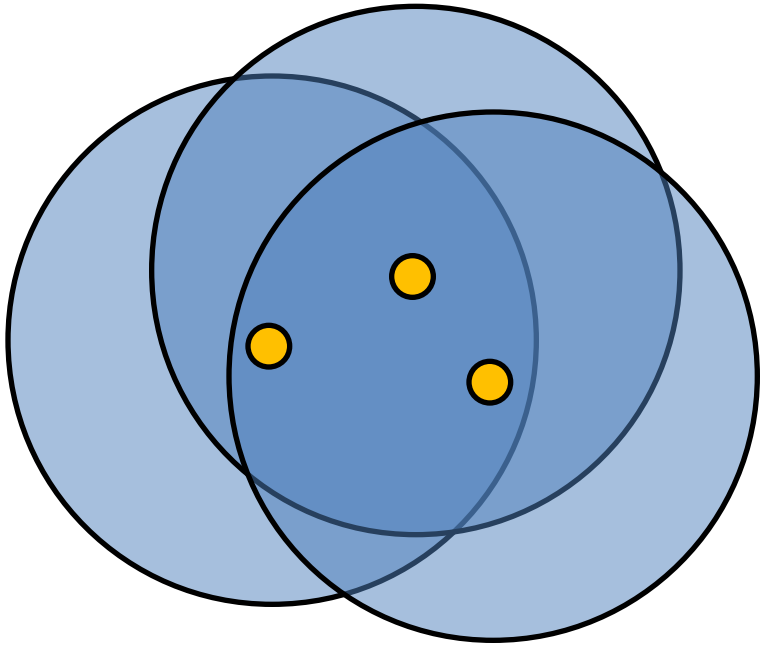


- RPL supports P2P traffic.
  - When node A sends a packet destined to node B, if B is not in direct reach, it forwards the packet to its DODAG parent.
  - From there, if the destination is reachable from one of its children, the packet is forwarded in the down direction.
  - In other words, the packet travels up to a common ancestor at which point it is forwarded in the down direction toward the destination.
- An interesting optimization periodically emits *link-local* multicast IPv6 DAOs -→ if the destination is in direct range (one hop away), a node can send the packet directly to the destination without following the DODAG.
- Sending DIO and DAO messages is governed by the use of trickle timers. The trickle timers use dynamic timers that govern the sending of RPL control messages in an attempt to reduce redundant messages.
  - When the DODAG is unstable (e.g., the DODAG is being rebuilt) RPL control messages are sent more frequently (the DODAG becomes inconsistent).
  - On the other hand, as the DODAG stabilizes messages are sent less often to reduce the control plane overhead, which is very important in LLNs.

skip

Trickle: idea

- Concerns
  - Broadcast is expensive
  - Wireless channel is a shared, spatial resource
- Idea
  - Dynamic adjustment of transmission period
  - Suppress transmissions that may be redundant
- Parameters:
  - T_min: Minimum advertisement period
  - T_max: Maximum advertisement period
  - k: Suppression threshold
- Period adjustment:
  - On receiving *inconsistent* route information, reset to T_min
  - Otherwise, double up to T_max
- Suppression:
  - Increment count when receiving *similar* advertisement
  - At end of period, transmit if c < k, set c = 0
- Proposal:
  - Carry T_min, T_max, and k in DIO

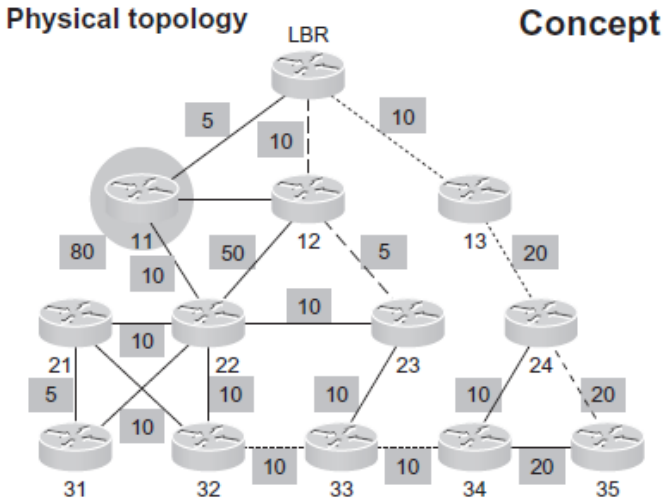


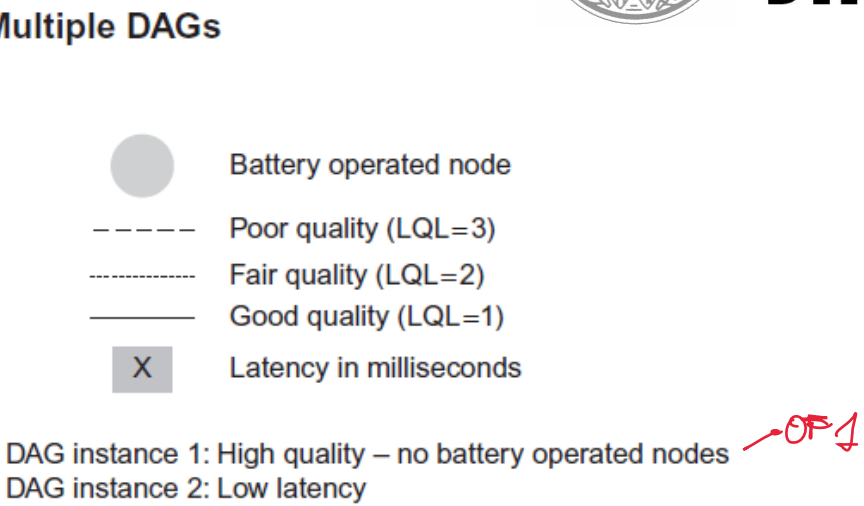
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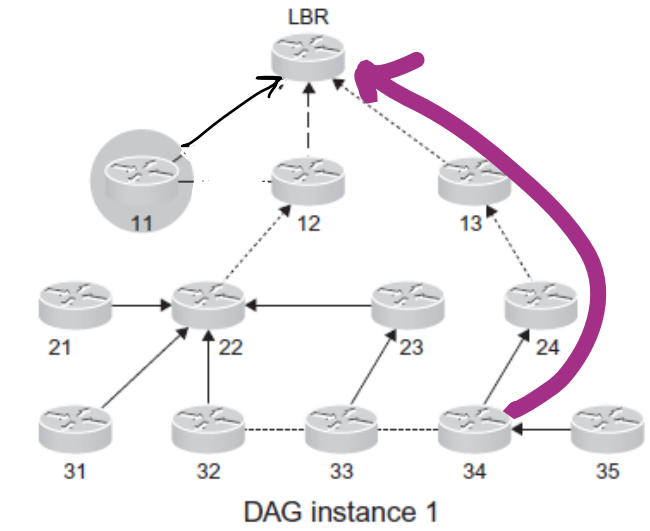
RPL Operations (5/5)

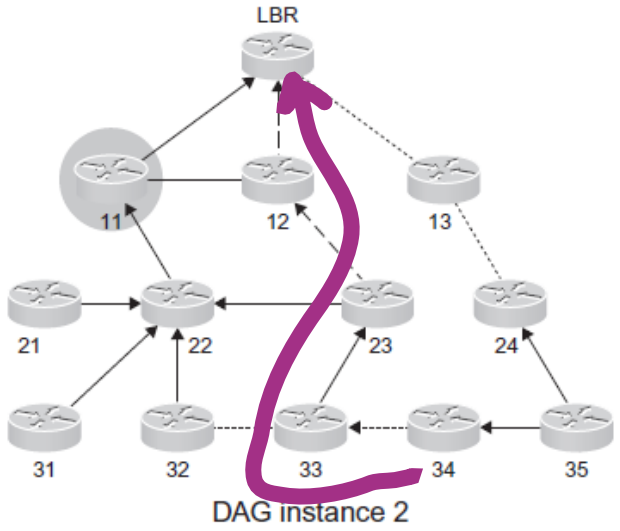
- Once the DODAG is built and routing tables are populated, routing is fully operational.
- As links and nodes fail, paths are repaired using local and global repair mechanisms.
  - Local repairs quickly fi nd a backup path without an attempt to globally reoptimize the DODAG entirely
  - Global repairs rely on a reoptimization process driven by the DODAG root.
- RPL also supports the concept of DODAG instances identifi ed by an Instance ID called the RPLInstanceID. It might be useful to form different topologies according to various sets of constraints and OFs. An RPL node may join multiple DODAG instances; for example:
  - one DODAG optimizes for high reliability
  - another DODAG optimizes for low latency.
- Data packets are then forwarded along the appropriate DODAG according to the application requirements.

Multiple DODAG Instances





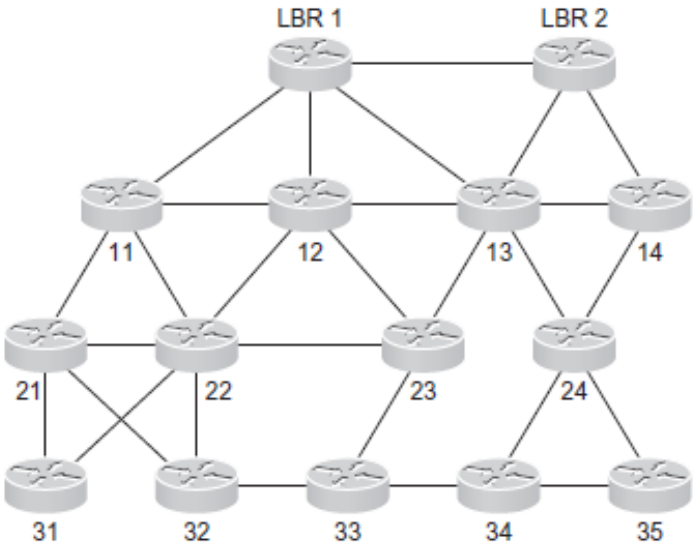




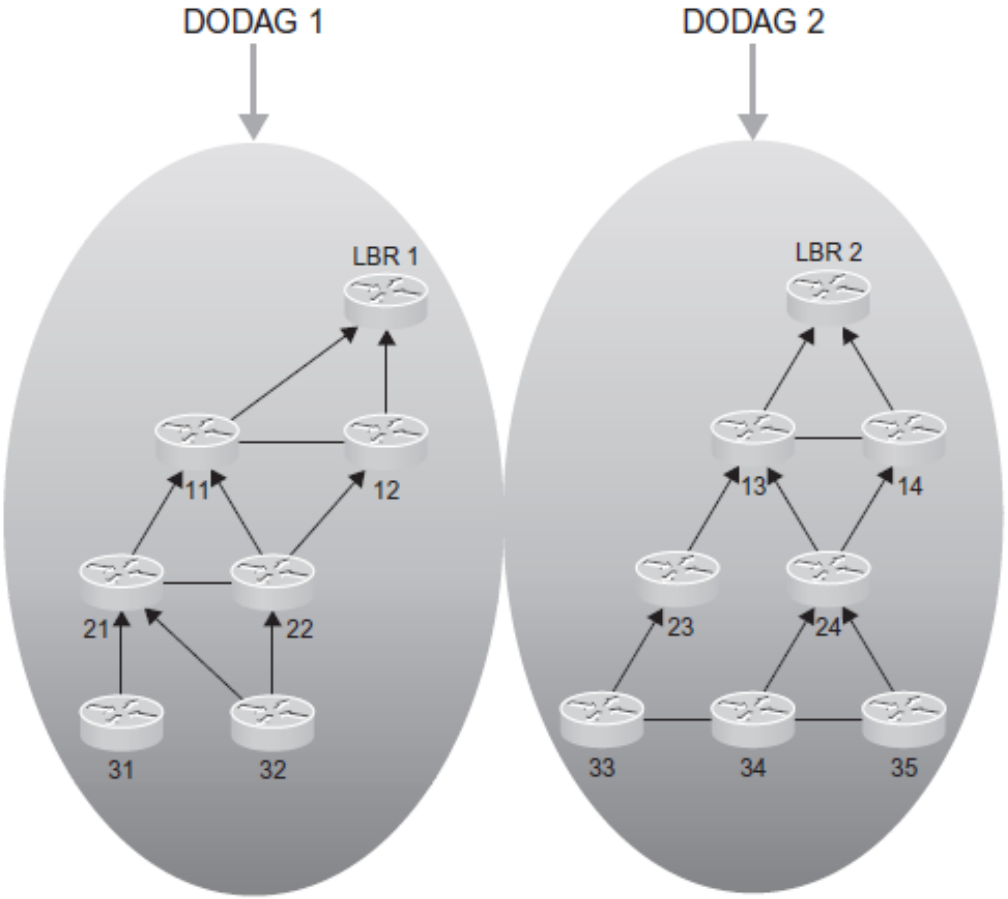
Multiple Disjoint DODAGs

Not necessarily

Physical topology



Multiple DODAGs



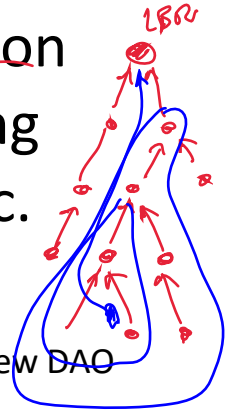
DIO Messages





- DIO messages are sent by RPL nodes to advertise a DODAG and its characteristics → DIOs are used for DODAG discovery, formation, and maintenance.
- DIOs carry a set of mandatory information augmented with options.
- The following flags and fields are currently defined:
  - Grounded (G): Indicates whether the DODAG is grounded, in other words, the DODAG root is a goal for the OF (e.g., the DODAG root is connected to a non-LLN IP network such as a private network or the public Internet).
  - Destination Advertisement Trigger (T): The T bit is used to trigger a complete refresh of the routes in the down direction (downward routes).
  - Destination Advertisement Stored (S): The S bit is used to indicate that a non-root ancestor is storing routing table entries learned from DAO messages.
  - Destination advertisement supported (A flag): The A flag is set when the DODAG root supports the collection of prefix advertisements and enables the advertisement of prefixes in the DODAG.
  - DODAGPreference (Prf): The Prf is a 3-bit field set by the DODAG root to report its preference. It can be used to engineer the network and make some DODAGs more attractive to join.
- The DODAGSequenceNumber is the sequence number of the DODAG that characterizes the DODAG iteration and is exclusively controlled by the DODAG root.
- The RPLInstanceID is used to identify the DODAG instance and is provisioned at the DODAG root.
- The Destination Advertisement Trigger Sequence Number (DTSN) is an 8-bit integer set by the node sending the DIO → used to maintain the downward routes.
- The DODAGID is a 128-bit integer set by the DODAG root and that uniquely identifies the DODAG.
- The DODAG Rank is the rank of the node sending the DIO message
  - The rank determines the relative position of a node in the DODAG and is used primarily for *loop avoidance*.
  - The rank is computed according to the OF and is potentially subject to local node policy.
  - The rank (although potentially derived from routing metrics) is not a metric.
- When two nodes have the same rank, the nodes are said to be siblings (they are located at a similar level of optimality in the DODAG).

DAO Messages



- DAO messages are used to propagate destination information along the DODAG in the up direction to populate the routing tables of ancestor nodes in support of P2MP and P2P traffic.
- The DAO message includes the following information:
  - DAO sequence: A counter incremented by the node owning the advertised prefix each time a new DAO message is sent.
  - RPLInstanceID: The topology instance ID as learned from the DIO.
  - DAO rank: Corresponds to the rank of the node that owns the prefix.
  - DAO lifetime: It is expressed in seconds and corresponds to the prefix lifetime.
  - Route tag: 8-bit integer that can be used to tag " critical " routes. The priority could be used to indicate whether the route should be stored by the nodes with a lower rank (closer to the DODAG root), which could be useful if nodes have limited memory capacities and must be selective about which destination information to cache. Note that the size of that fi eld has been changed several times and is subject to further changes.
  - Destination Prefix: The Prefix Length field contains the number of valid leading bits in the prefix.
  - Reverse Route Stack (RRS): contains a number of RRCount (another field of the DAO message) IPv6 addresses used in LLNs with nodes that cannot store routing tables.

The "perfect" DAG?





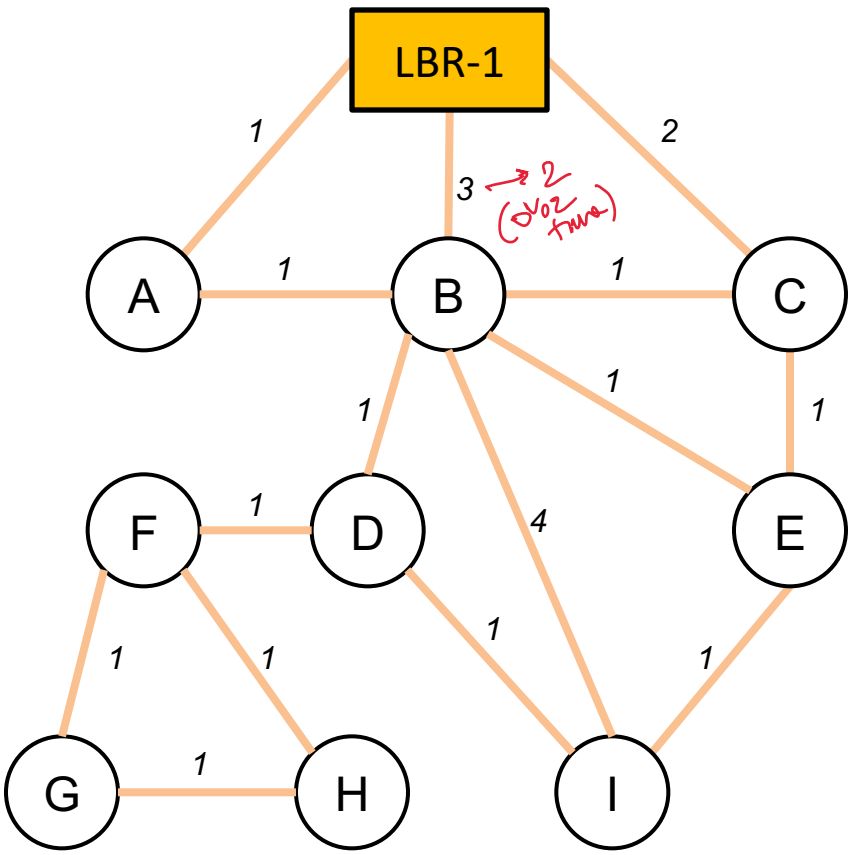
- Since there no perfect wires with constant properties there's hardly a perfect DAG
- Conditions change, batteries deplete
- Need to arbiter between conflicting metrics and goals
- Build a structure that is satisfactory for all
- Maintain with probably a slow evolution
- The characteristics of the network, including the metrics, are unpredictable. What we really want is that every node gets an acceptable result most of the time.
- There are many ways in which acceptable can be arranged (for some time).
- We do not look for the illusory "perfect DAG" so we could reach any of a large number of acceptable solutions.

An Open Issue: Carrying IPv6 Addresses



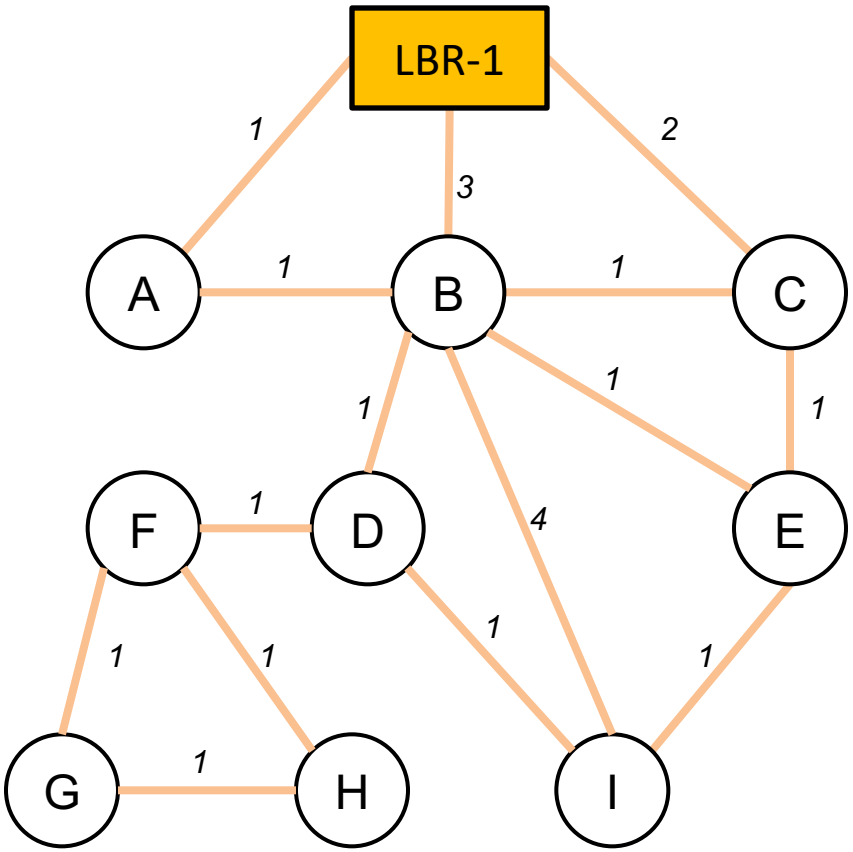
- High communication overhead
  - Carried in DIO, DAO, source routes
- High state overhead
  - Maintain destination routing state
- Options for compressing
  - Use a new namespace (e.g. labels)
  - Compress IPv6 addresses

DAG Construction



- LLN links are depicted
- Links are annotated w/ ETX
- It is expected that ETX variations will be averaged/filtered to be stable enough for route computation
  - Nodes are also to observe the metric and gain confidence before use

DAG Construction



- Example: Objective Code Point (default OF)
- Main characteristics
  - Metric: ETX
  - Objective: Minimize ETX
  - Depth computation:
    Depth ~ ETX
    - Note that a practical computation may be more coarse

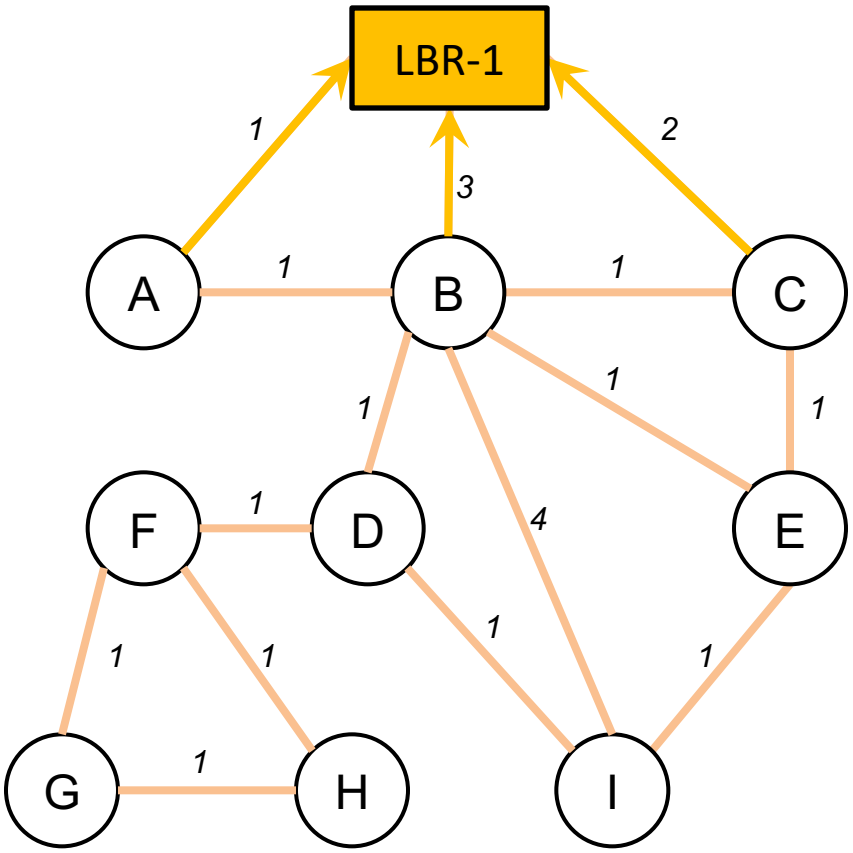
Global aggregated metric

$$ETX_{path} = \sum_{i=1}^{n} ETX_{link_i}$$

Links over the path.

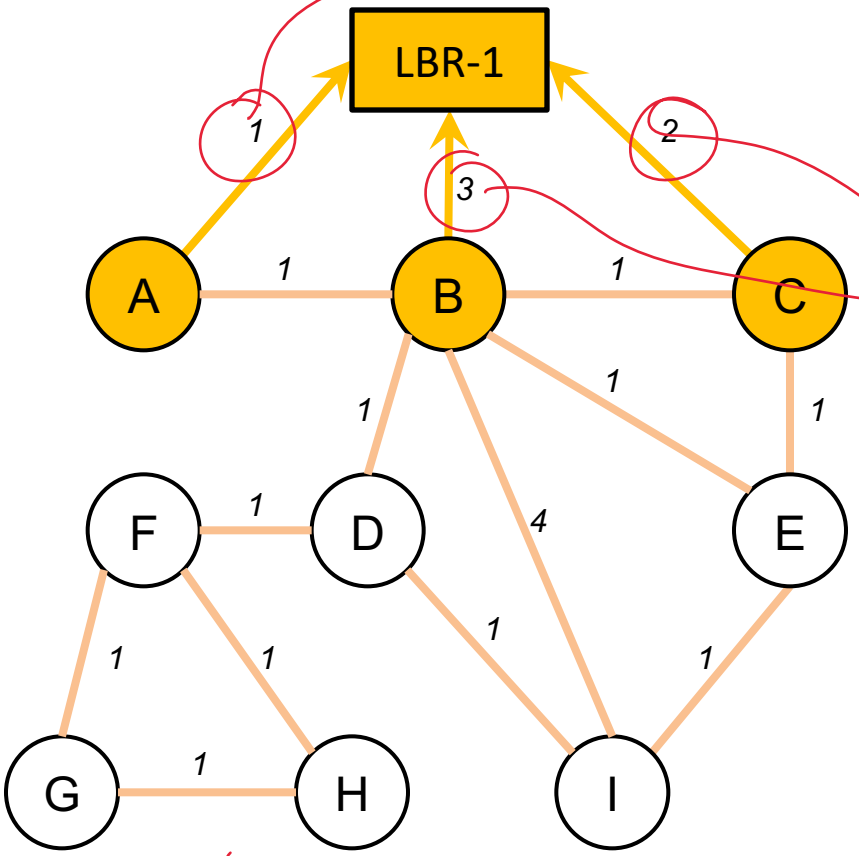
$\{link_i\}_{i=1}^{n}$

DAG Construction

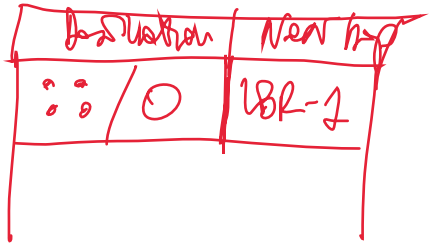


- LBR-1 multicasts RA-DIO
- Nodes A, B, C receive and process RA-DIO
- Nodes A, B, C consider link metrics to LBR-1 and the optimization objective
- The optimization objective can be satisfied by joining the DAG rooted at LBR-1
- Nodes A, B, C add LBR-1 as a DAG parent and join the DAG

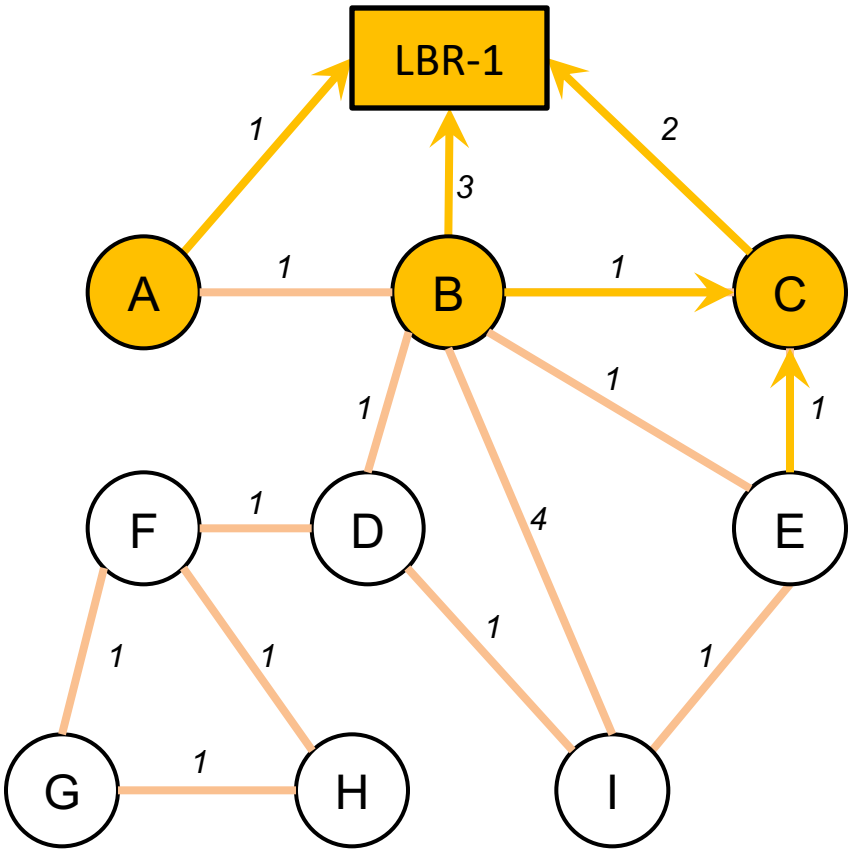
DAG Construction



- Node A is at Depth 1 in the DAG, as calculated by the routine indicated by the example OCP (Depth ~ ETX)
- Node B is at Depth 3, Node C is at Depth 2
- Nodes A, B, C have installed default routes (::/0) with LBR-1 as successor

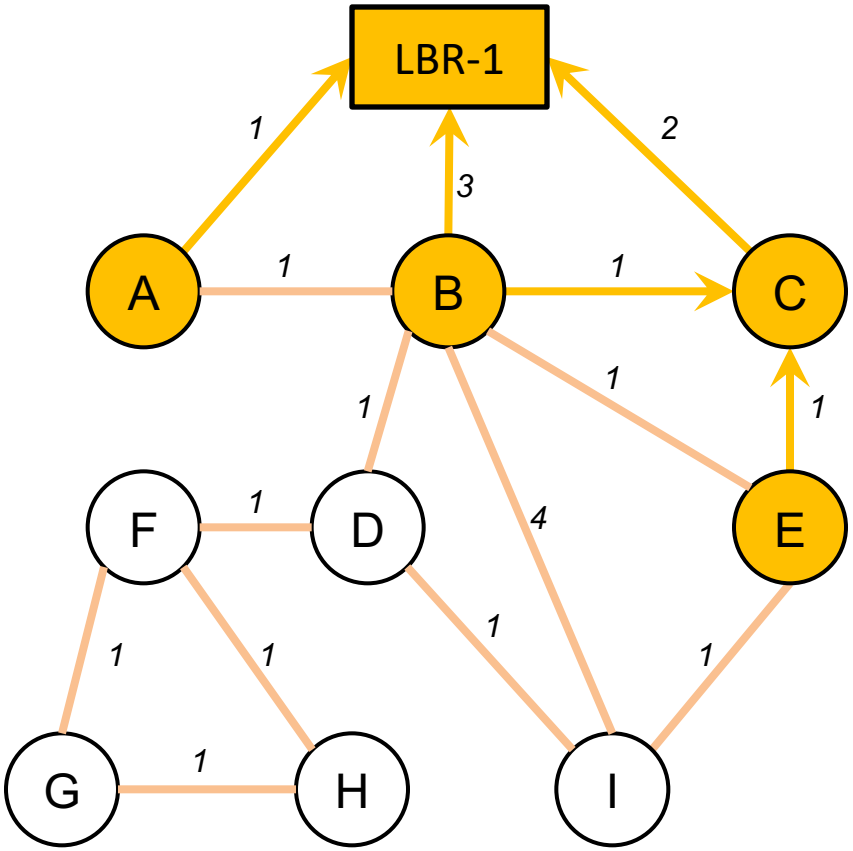


DAG Construction



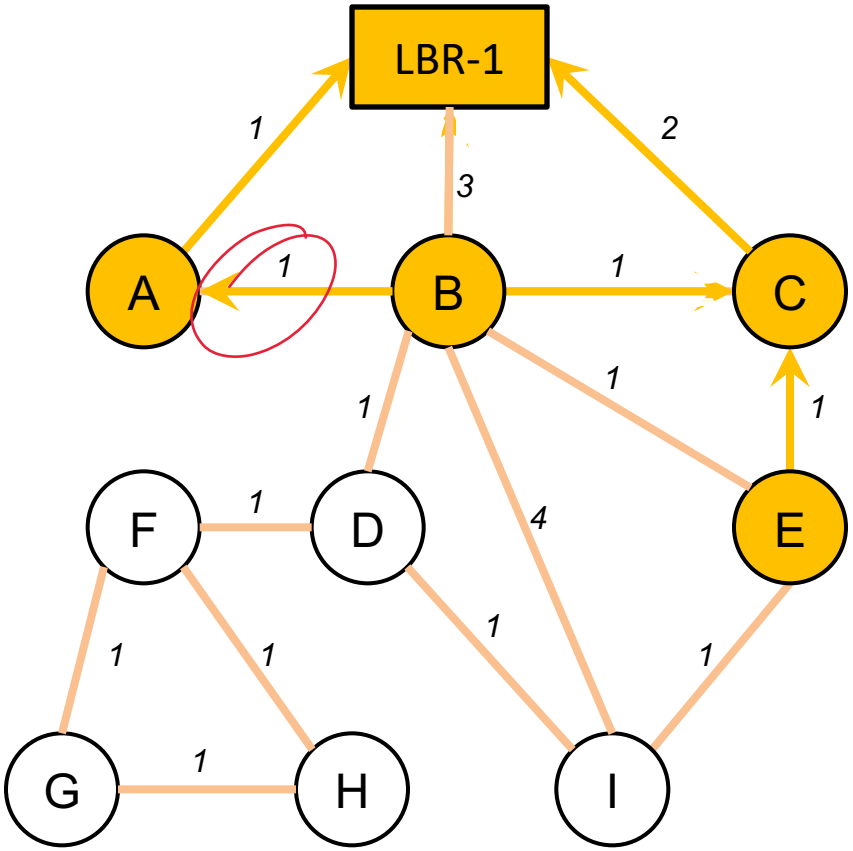
- The RA (Resource Advertisement) timer on Node C expires first (before A and B)
- Node C multicasts RA-DIO
- LBR-1 ignores RA-DIO from deeper node (DIO → there will be the rank=2 of C)
- Node B can add Node C as *alternate* DAG Parent, remaining at Depth 3
- Node E joins the DAG at Depth 3 by adding Node C as DAG Parent

DAG Construction



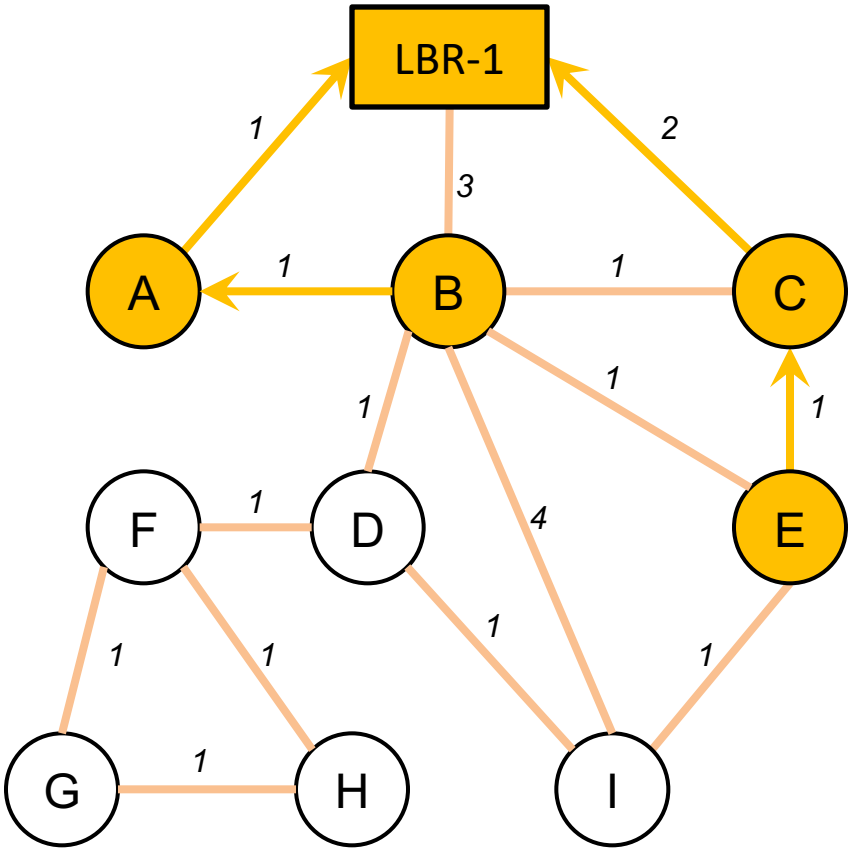
- Node A is at Depth 1, and can reach ::/0 via LBR-1 with ETX 1
- Node B is at Depth 3, with DAG Parents LBR-1, and can reach ::/0 via LBR-1 or C with ETX 3
- Node C is at Depth 2, ::/0 via LBR-1 with ETX 2
- Node E is at Depth 3, ::/0 via C with ETX 3

DAG Construction



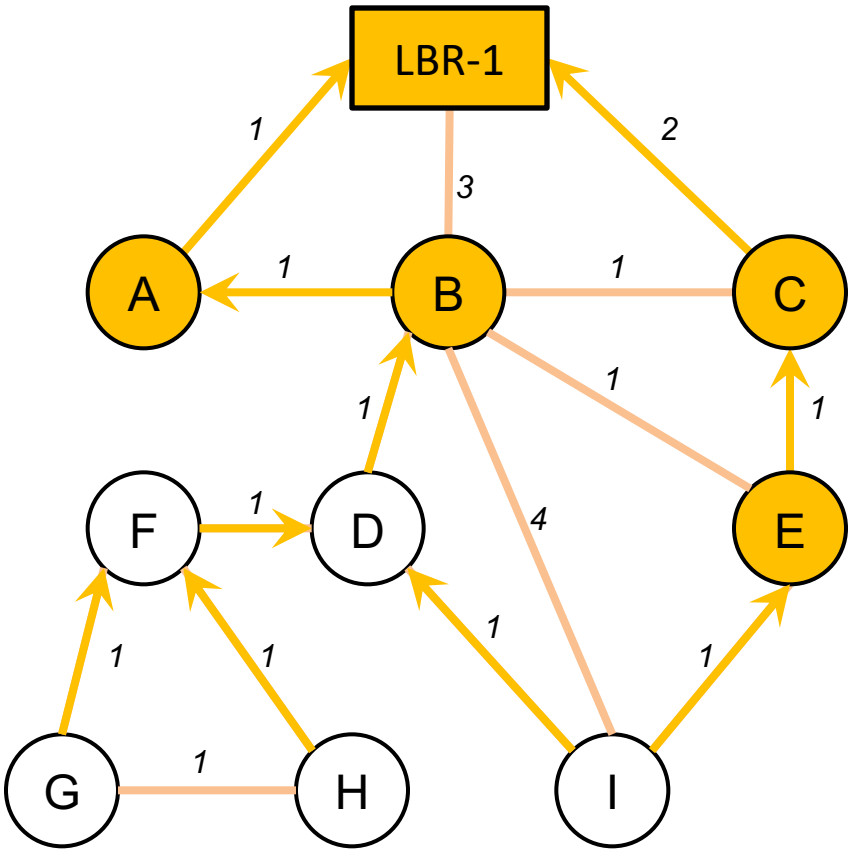
- The RA timer on Node A expires
- Node A multicasts RA-DIO
- LBR-1 ignores RA-DIO from deeper node
- Node B adds Node A
- Node B can improve to a more optimum position in the DAG
- Node B *removes* LBR-1, Node C as DAG Parents

DAG Construction

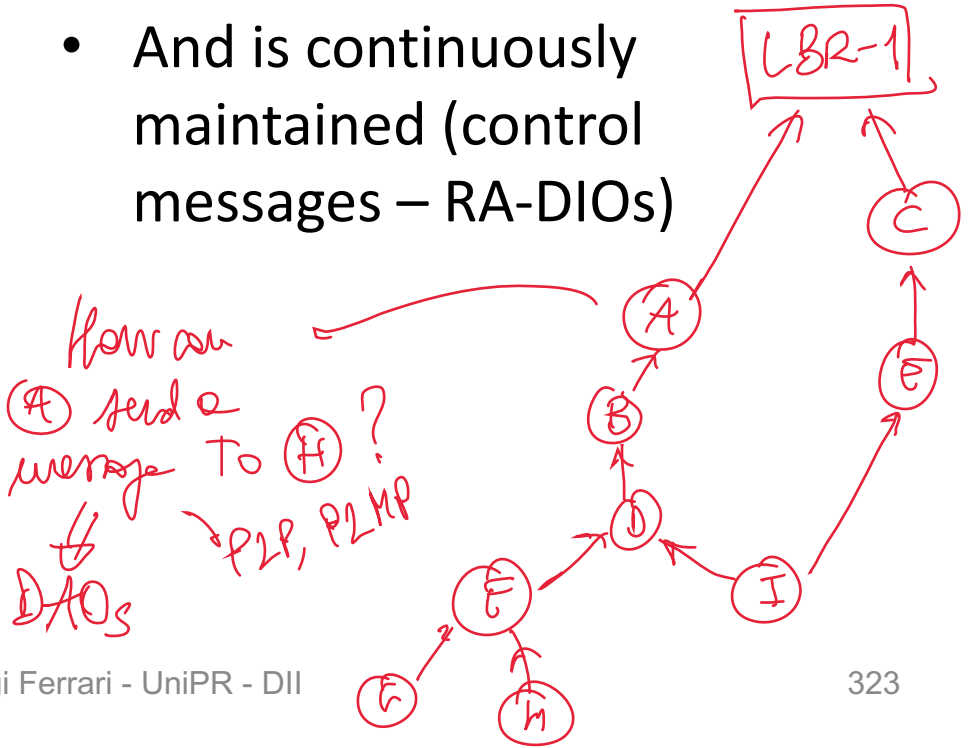


- Node A is at Depth 1, ::/0 via LBR-1 with ETX 1
- Node B is at Depth 2, ::/0 via A with ETX 2
- Node C is at Depth 2, ::/0 via LBR-1 with ETX 2
- Node E is at Depth 3, ::/0 via C with ETX 3

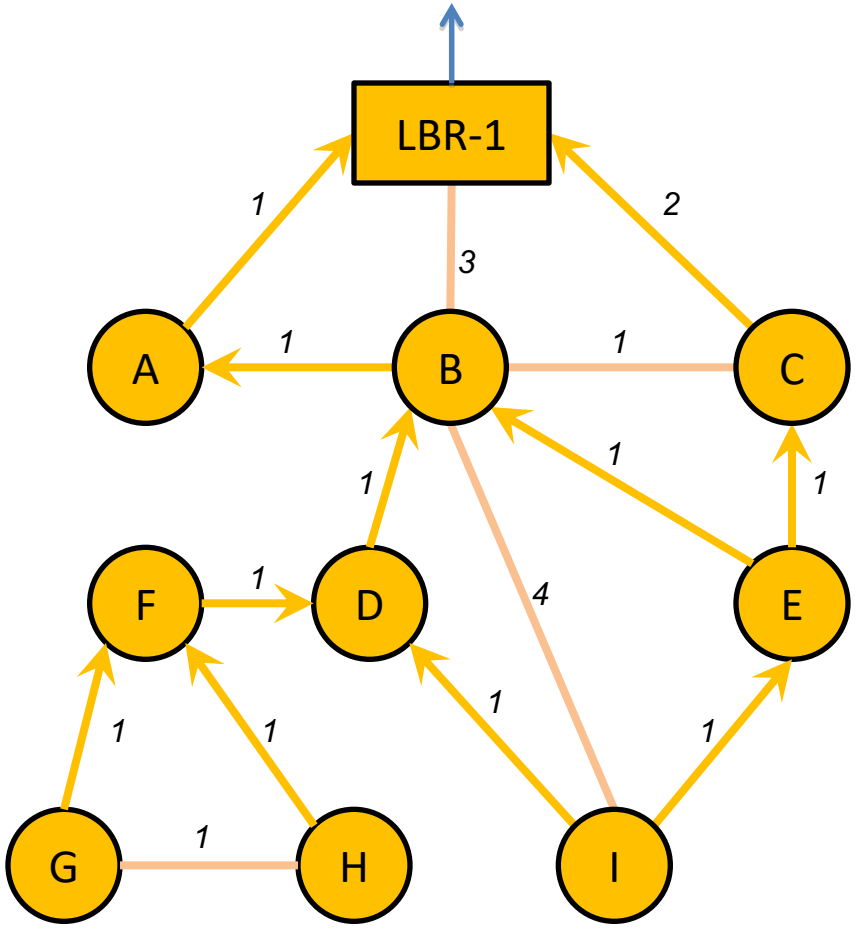
DAG Construction



- DAG Construction continues…
- And is continuously maintained (control messages – RA-DIOs)

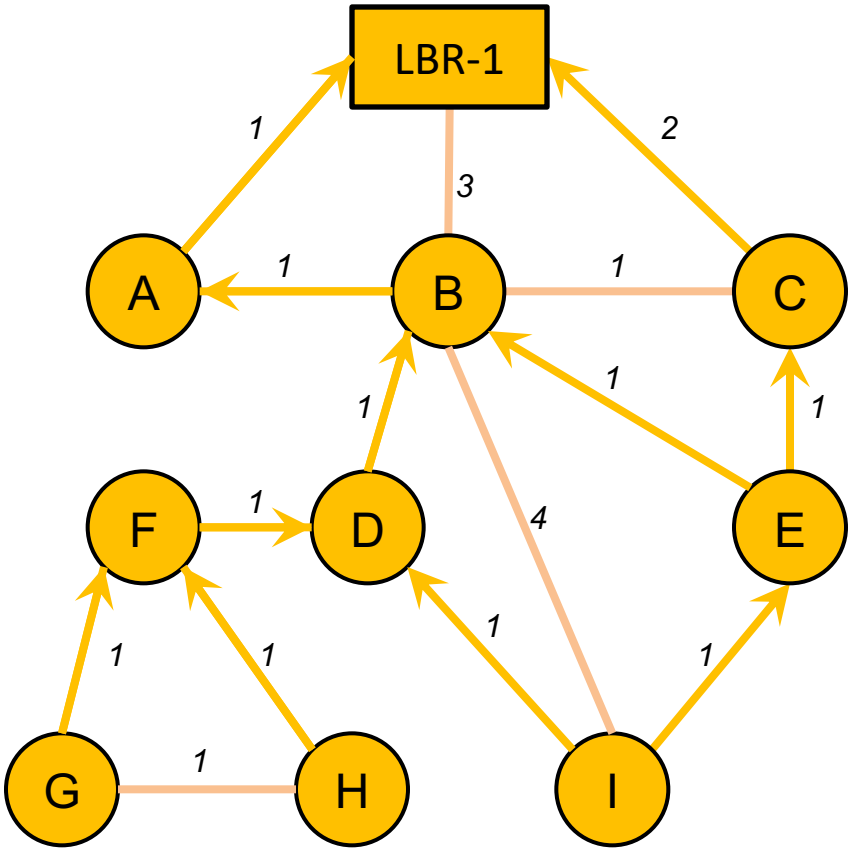


MP2P Traffic



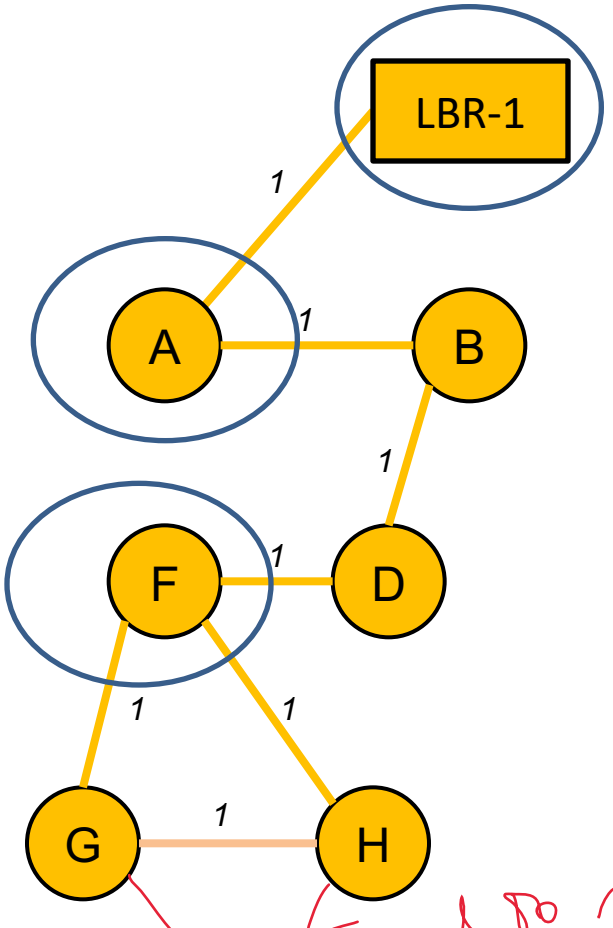
- MP2P traffic flows upwards along DAG, toward DAG Root
- DAG Root may also extend connectivity to other prefixes beyond the DAG root, as specified in the DIO
- Nodes may join multiple DAGs as necessary to satisfy application constraints

Destination Advertisements



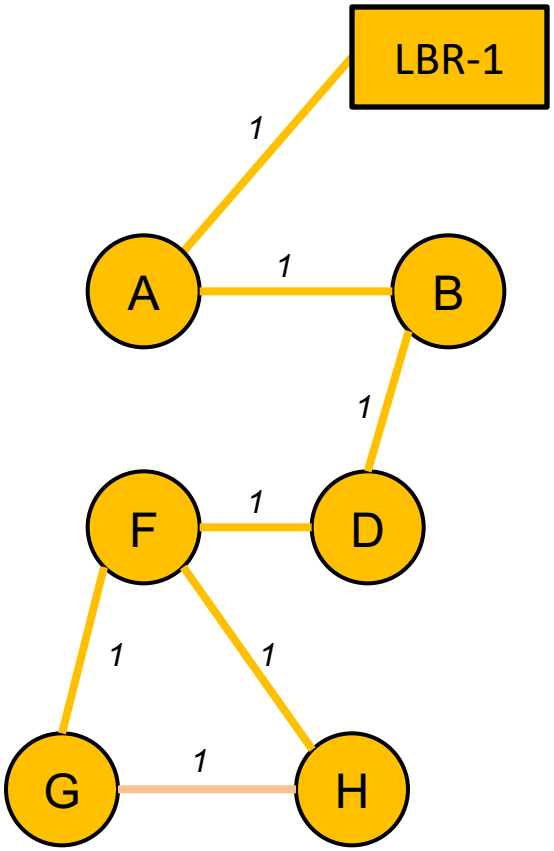
- Destination Advertisements (Das) build up routing state in support of P2MP traffic flows downward, from the sink to other nodes
- DAs use the same DAG
- For simplicity, we will focus on a subset of DAs in the example

Destination Advertisements



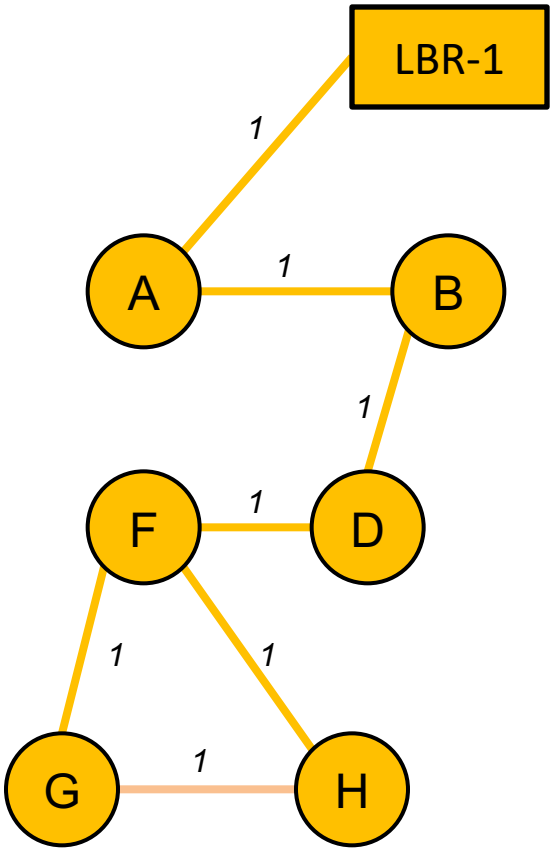
- Some nodes may be able to store routing state for downward flows (LBR-1, A, F)
- Some nodes may not (B, D)
- Some nodes may have a limited ability; DAs may indicate a priority for storage

Destination Advertisements



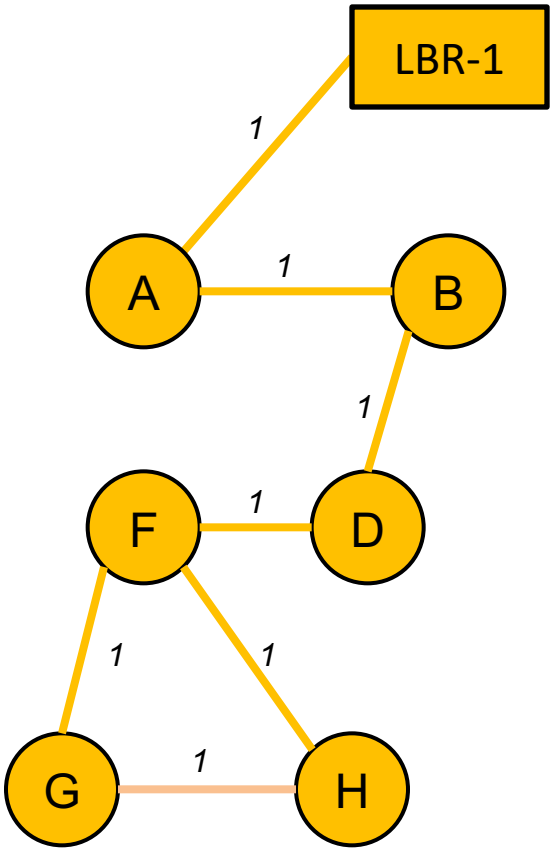
- DAs may be triggered by DAG root or node who detects a change
- DA timers configured such that DAs start at greater depth, and may aggregate as they move up

Destination Advertisements



- LBR-1 triggers Destination Advertisement mechanism in DIO
- G emits NA to F with DAO indicating reachability to destination prefix G::
- F stores G:: via G
- H emits NA to F for destination prefix H::
- F stores H:: via H

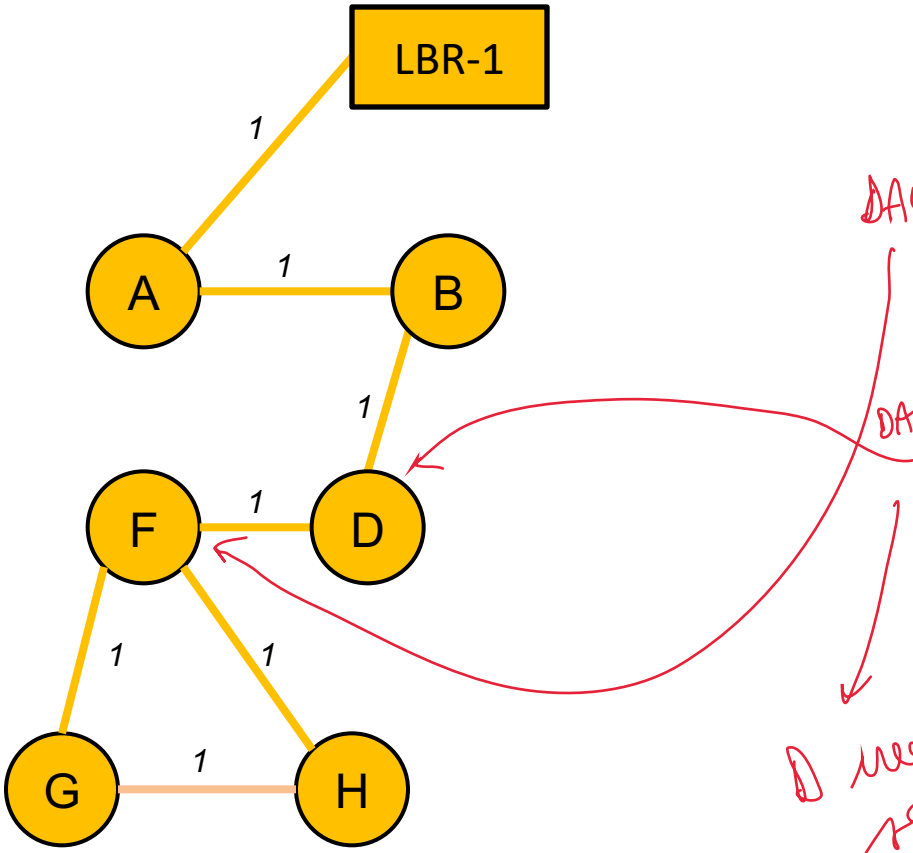
Destination Advertisements



- Suppose in this example F has a prefix F*:: capable of aggregating {F::, G::, H::}
  - The method to provision such a prefix is beyond the scope of RPL
- F emits NA to D with DAO indicating reachability to destination prefix F*::
- D cannot store…

(continued)

Destination Advertisements

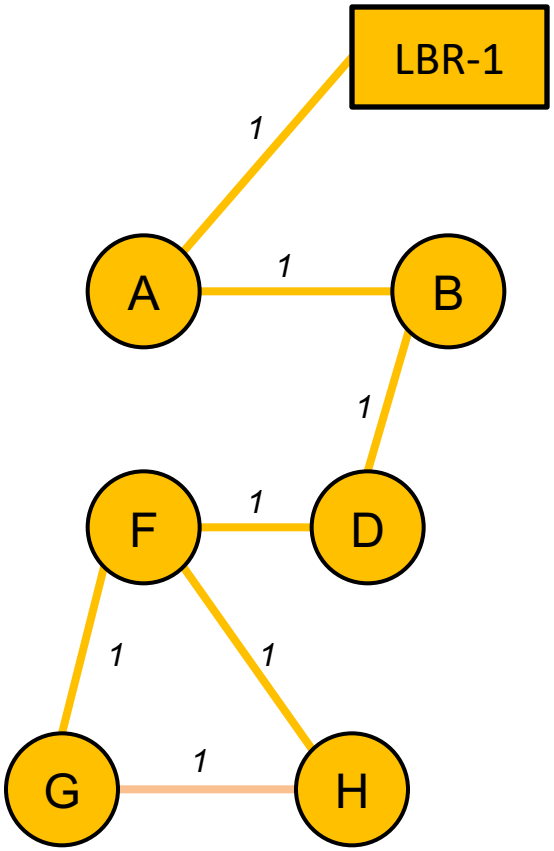


- D adds F to the Reverse Route Stack in the DAO, and passes DAO on to B for F*:: [F]
- D also emits a DAO indicating prefix D:: to B
- B cannot store routing state…

D needs to send 2 DAOs, because the prefix D:: and F*:: cannot be combined.

(continued)

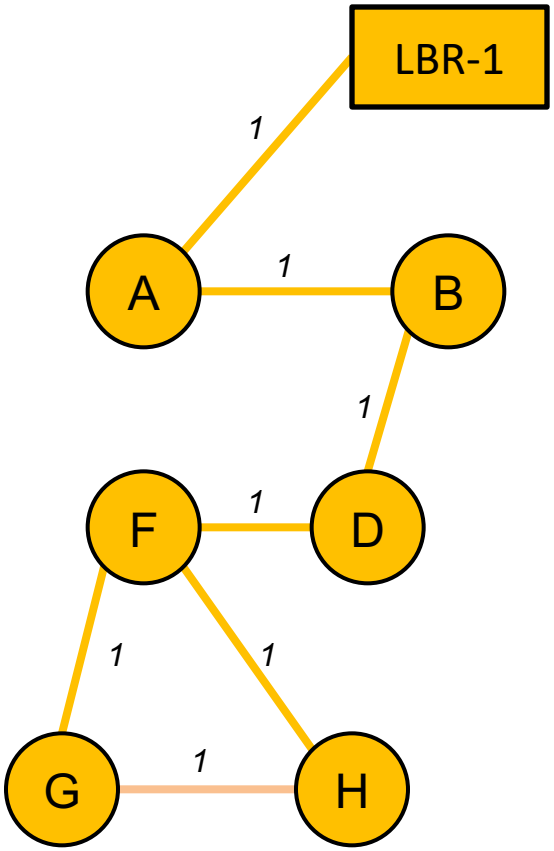
Destination Advertisements



- B adds D to the Reverse Route Stack in the DAO for D::, and passes DAO D:: [D] on to A
- A stores D:: via B, with the piecewise source route [D]
- B also emits a DAO indicating prefix B:: to A
- A stores B:: via B

(continued)

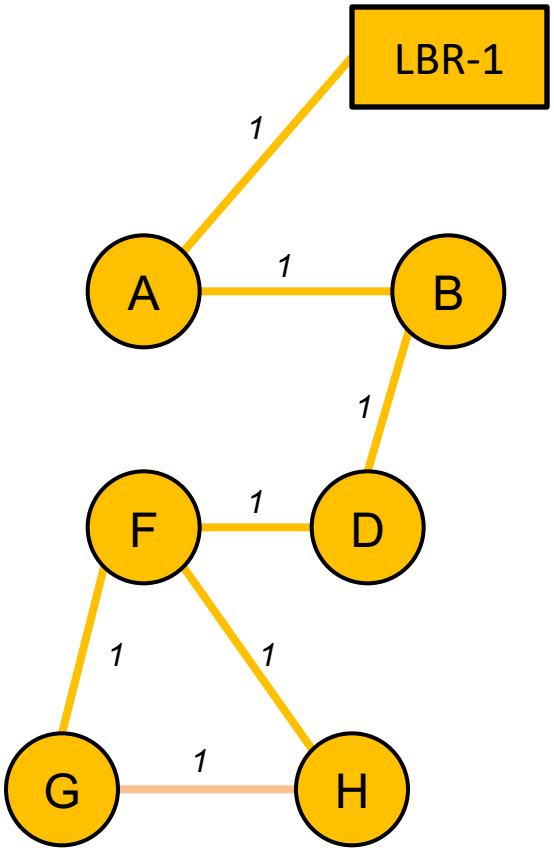
Destination Advertisements



- A emits DAOs to LBR-1 for destination prefixes A::, B::, D::, and F*
- LBR-1 stores A:: via A, B:: via A, D:: via A, and F*:: via A
- A stored B:: via B, D:: via B [D], F* via B [D, F]
- B, D stored nothing
- F stored G:: via G, H:: via H

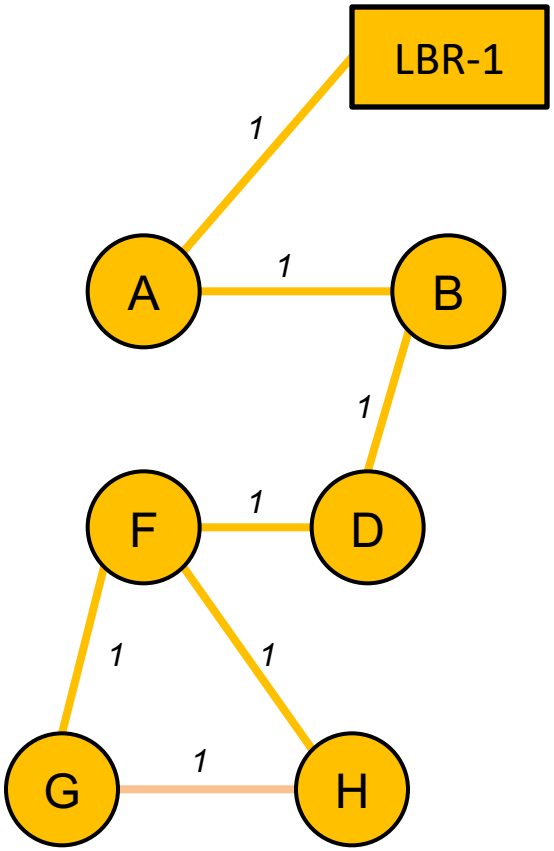
With DAOs, all nodes become aware of the structure of the network

P2MP Traffic



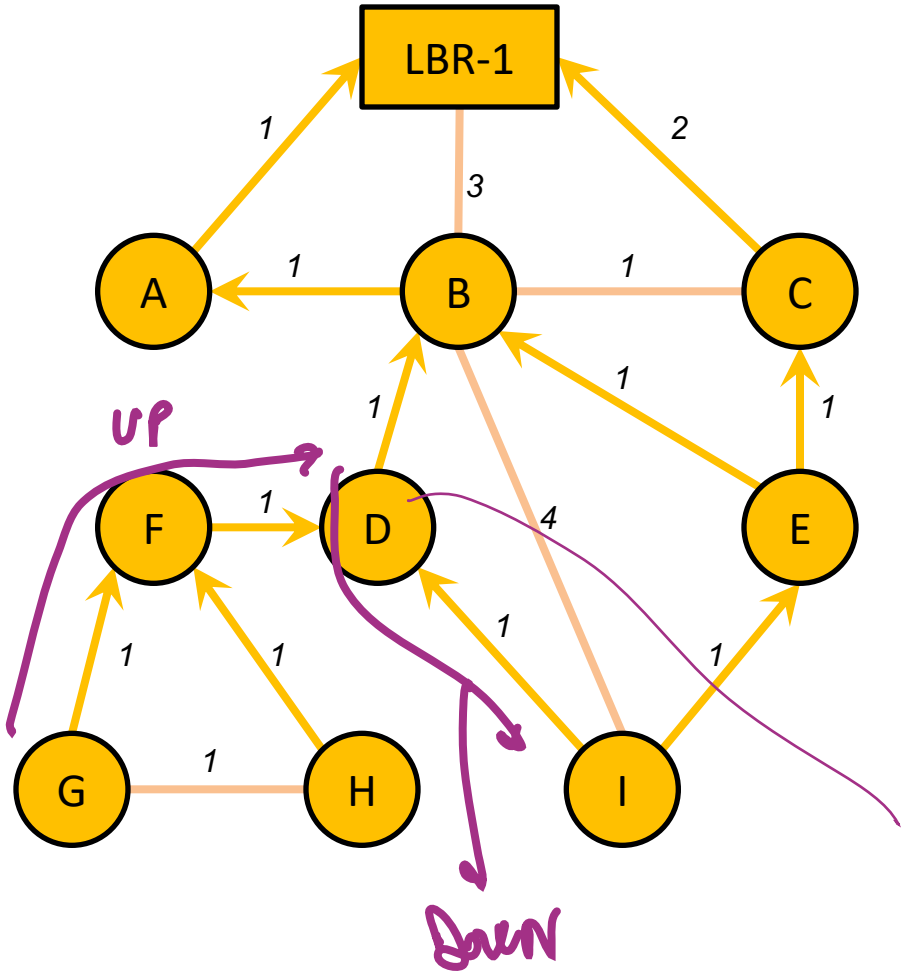
- The routing state setup by Destination Advertisement is used to direct P2MP traffic downward
- LBR-1 directs traffic for G (F*::) to A
- A adds source routing directive, [D, F], and forwards to B
- B uses source routing directive to forward to D...

P2MP Traffic



- D uses source routing directive to forward to F
- F uses routing state to forward to G
- Note the use of source routing to traverse the stateless region of the LLN
  - The details of the source routing mechanism are still to be defined

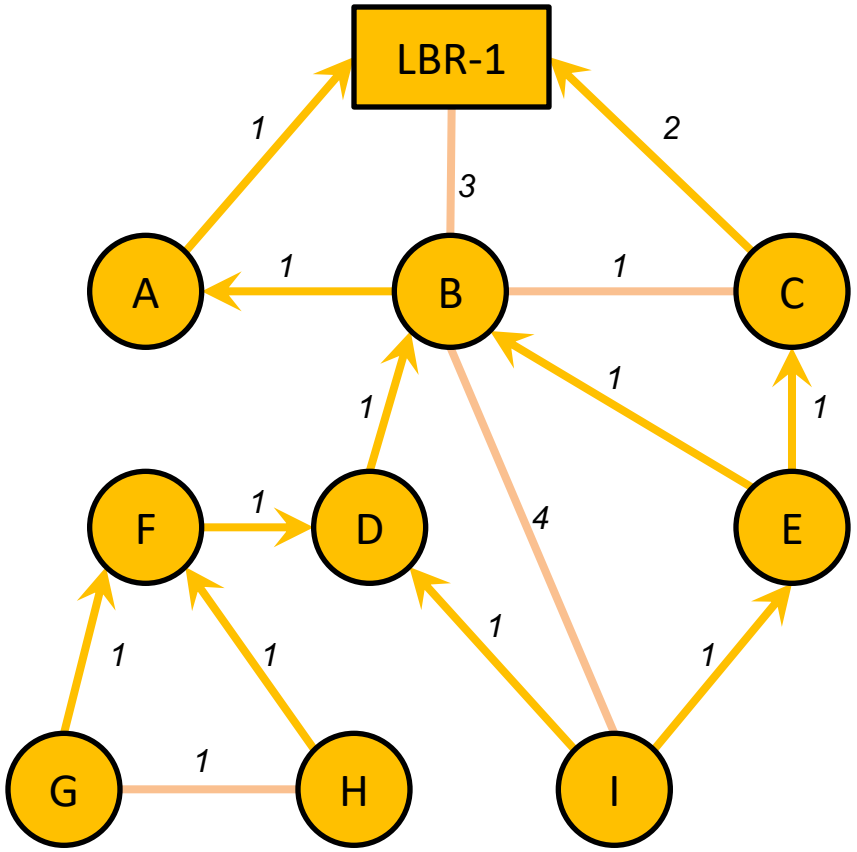
P2P Traffic



- By default, P2P traffic follows the default route until a node is encountered who has a more explicit route
- P2P traffic flows upwards along the DAG until a common node is reached to direct the traffic outwards

You start going down at the first node which knows a downward path to the destination

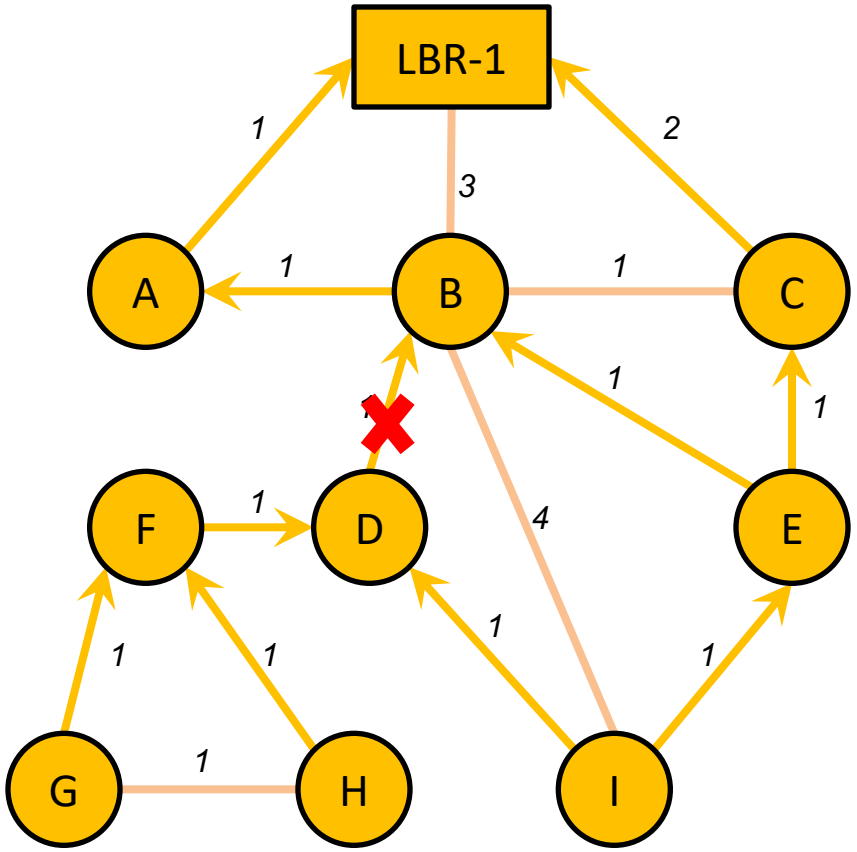
P2P Traffic



- The default scheme allows P2P traffic with minimal state required of nodes – but is not appropriate in all cases
- More optimal P2P routes may be computed and installed if the application requires
  - RPL does not specify nor preclude such a mechanism at this time

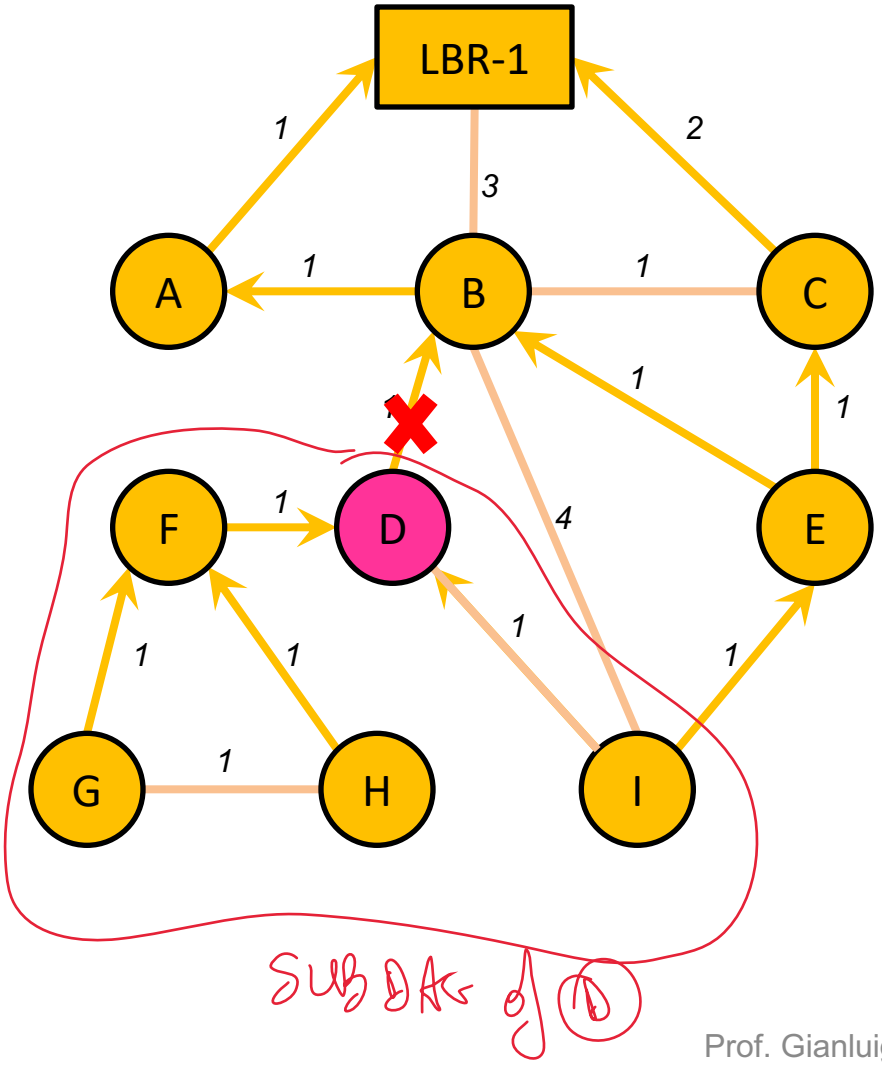
DAG Maintenance

Loop Avoidance



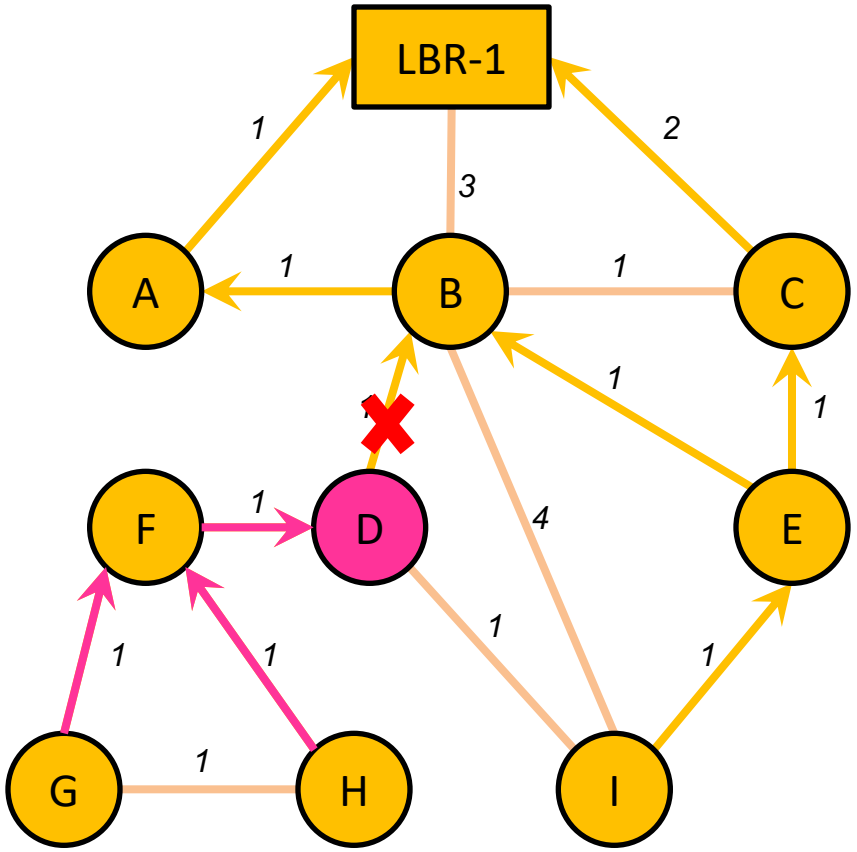
- Consider the case where the link B—D goes bad
- Node D will remove B from its DAG parent set
- Node D no longer has any DAG parents in the grounded DAG, so it will become the root of its own floating DAG

DAG Maintenance
Loop Avoidance



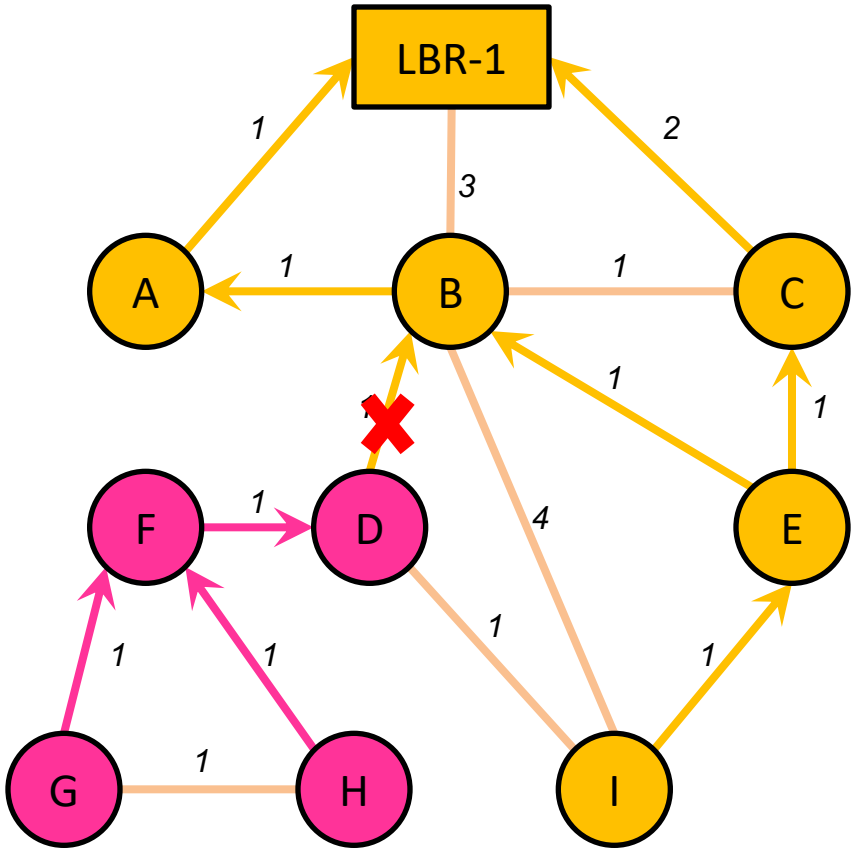
- Node D multicasts an RA-DIO to inform its sub-DAG of the change
- Node I has an alternate DAG Parent, E, and does not have to leave the DAG rooted at LBR-1.
- Node I removes Node D as a DAG Parent

DAG Maintenance
Loop Avoidance



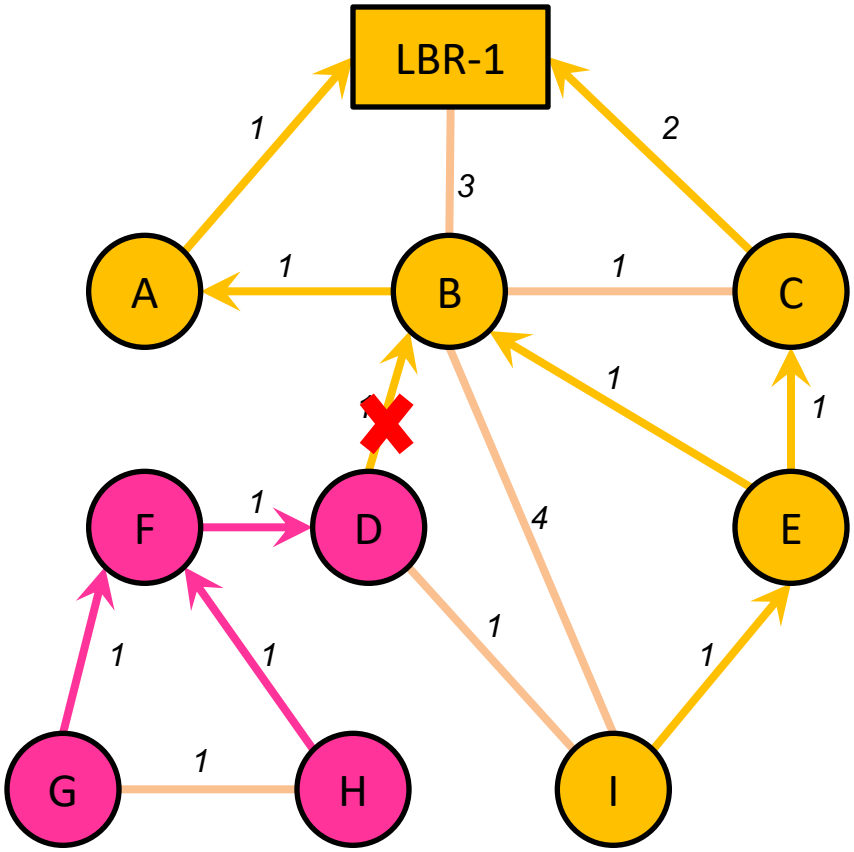
- Node F does not have an option to stay in the DAG rooted at LBR-1 (no alternate DAG Parents), so Node F follows Node D into the floating DAG
- Node F multicasts an RA-DIO
- Nodes G and H follow Node F into the floating DAG

DAG Maintenance
Loop Avoidance



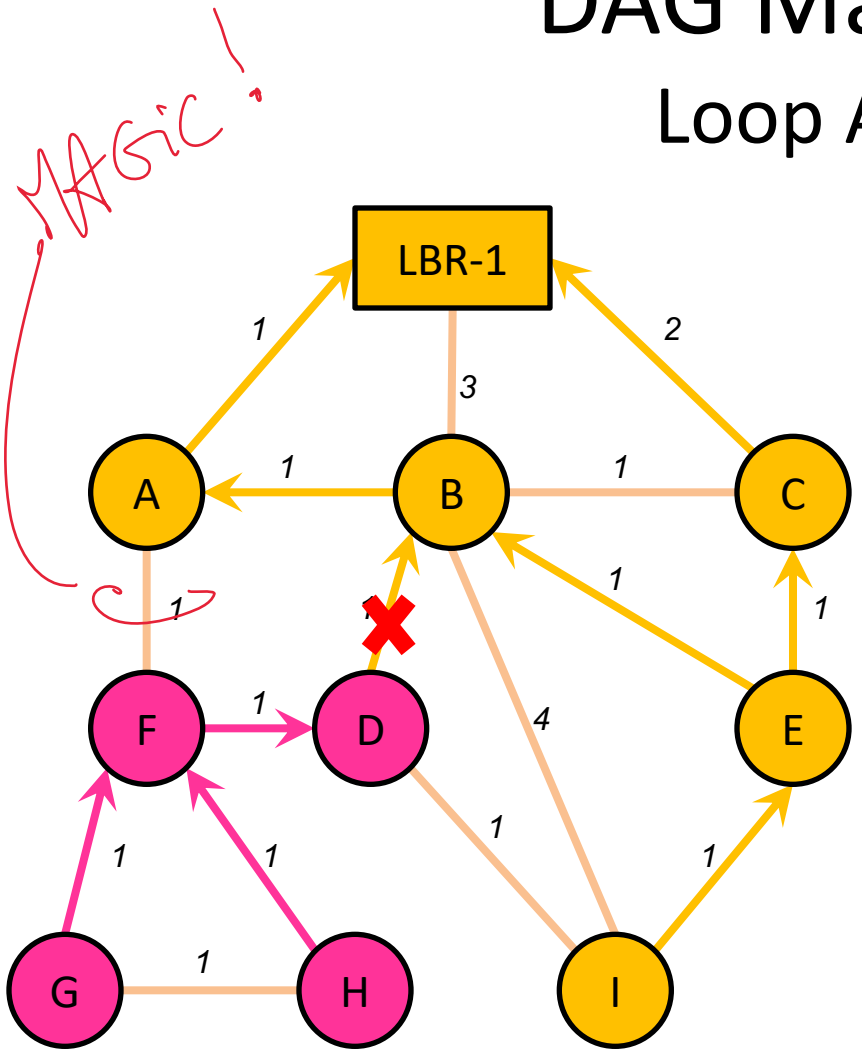
- The sub-DAG of node D has now been frozen
- Nodes contained in the sub-DAG have been identified, and by following node D into the floating DAG, all old routes to LBR-1 have been purged
- The floating DAG seeks to rejoin a grounded DAG…

DAG Maintenance
Loop Avoidance



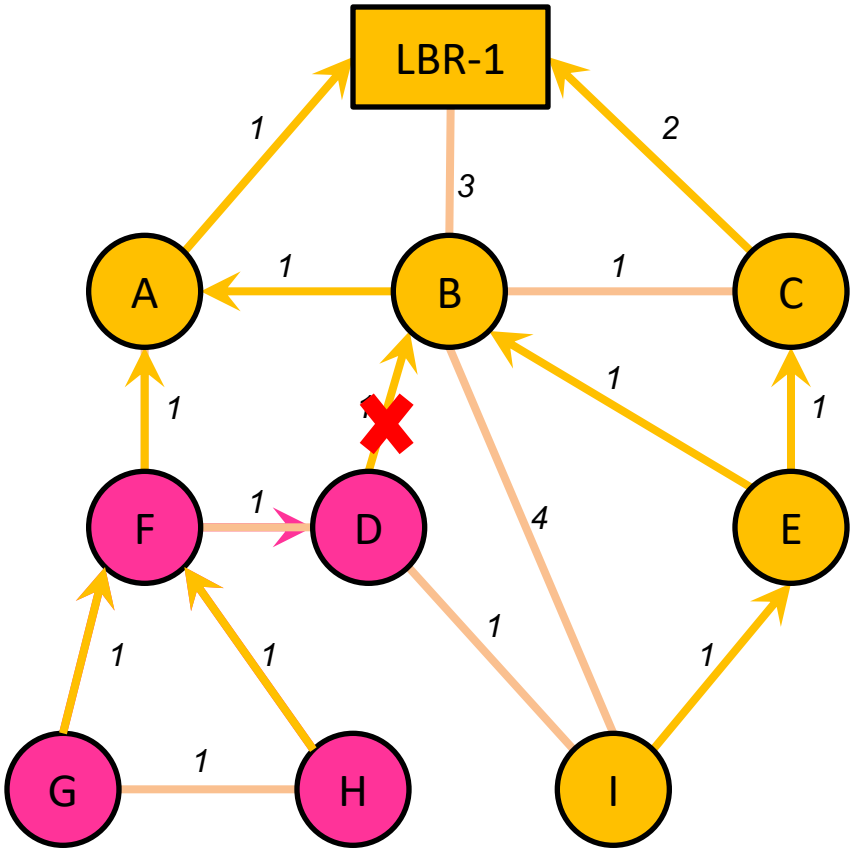
- Node I multicasts an RA-DIO
- Node D sees a chance to rejoin grounded DAG at depth 5 through Node I
- Node D starts a DAG Hop timer of duration α 4 associated with Node I

DAG Maintenance
Loop Avoidance



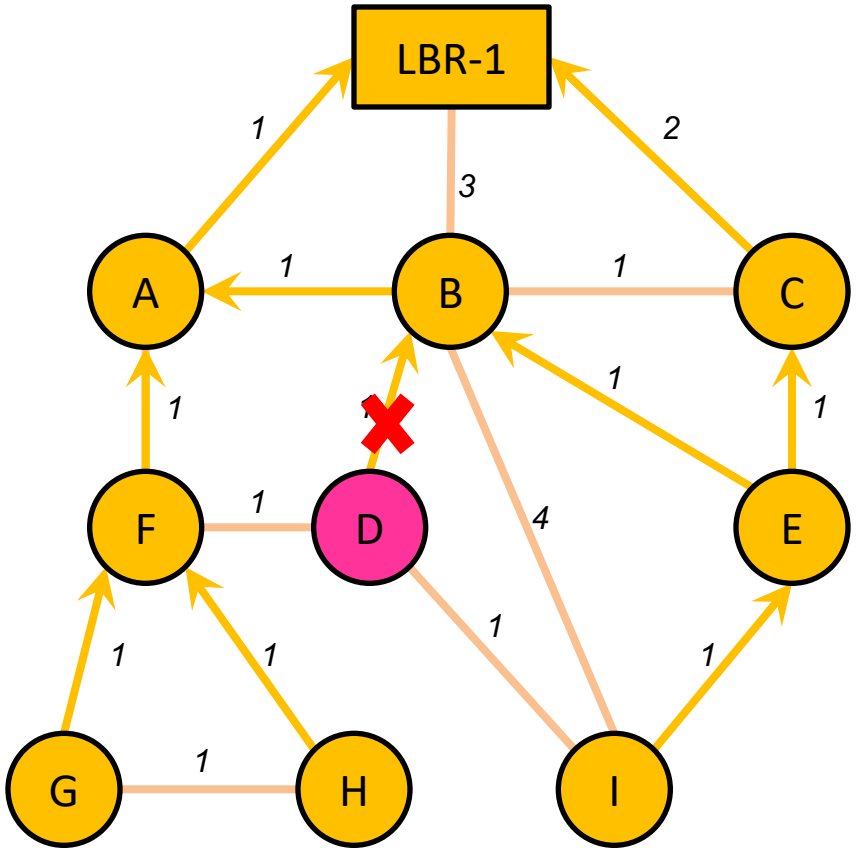
- Suppose a link A—F becomes viable
- Node A multicasts an RA-DIO
- Node F sees a chance to rejoin grounded DAG at depth 2 through Node A
- Node F starts a DAG Hop timer of duration α 1 associated with Node A

DAG Maintenance
Loop Avoidance



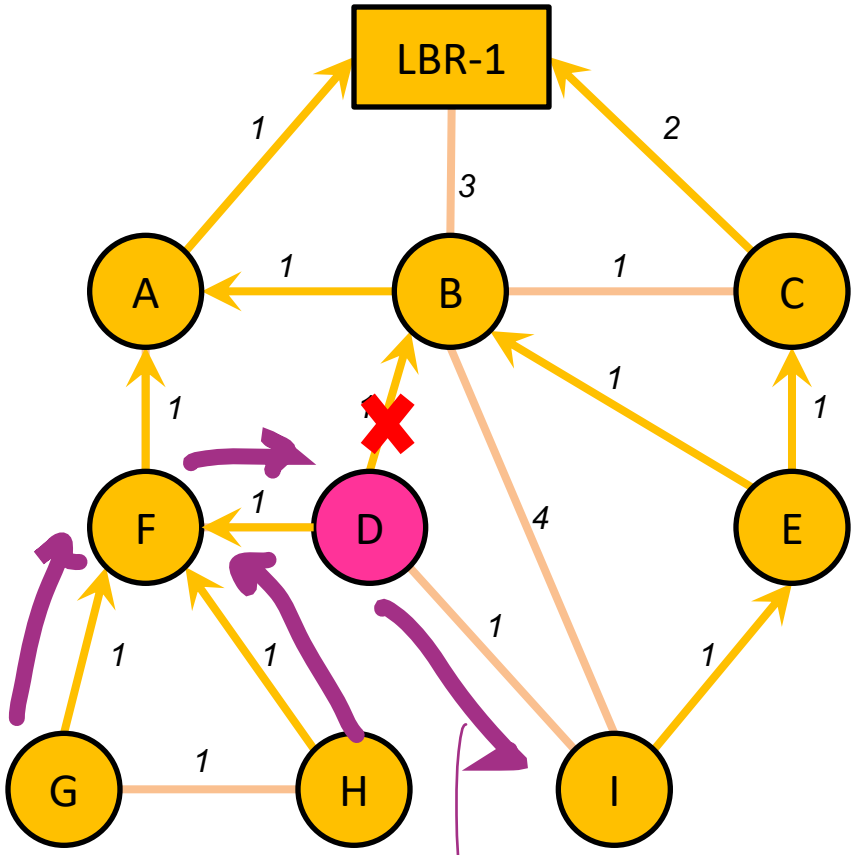
- Node F's DAG Hop Timer expires
- Node F joins to the grounded DAG at depth 2 by adding A as a DAG parent, and removing D
- Node F multicasts an RA-DIO
- Nodes G and H follow Node F to the grounded DAG

DAG Maintenance
Loop Avoidance



- Node D sees a chance to rejoin DAG LBR-1 at depth 3 through Node F
- Node D starts a DAG Hop timer of duration α 2 associated with Node F, in addition the DAG Hop timer already running with duration α 4 associated with Node I

DAG Maintenance
Loop Avoidance



- Node D's DAG Hop timer of duration α 2 tends to expire first
- Node D joins the grounded DAG at depth 3 by adding Node F as a DAG Parent
- The breaking-off and re-joining of the broken sub-DAG is thus coordinated with loop avoidance

Structure if the link A-F had not appeared